

# DECISION MAKING IN PARKINSONS'



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# INTRODUCTION



- Dopaminergic therapy in Parkinson's disease can alter reward processing and increase risk-taking behavior.
- Understanding how patients evaluate rewards, risks, and feedback is important for studying decision-making mechanisms

## GOAL:

- Characterize risk-taking behavior in Parkinson's patients.
- Investigate how feedback affects subsequent decisions.
- Explore neural correlates of decision-making using STN local field potential recordings.

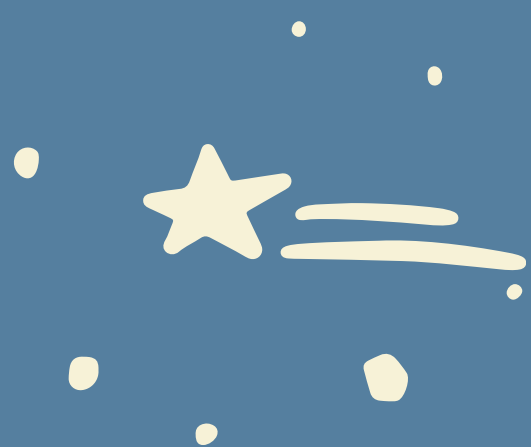


# TASK DESCRIPTION



## **Reinforcement Learning Gambling Task**

- Patients performed the task after 12 hours of medication withdrawal
- Patients were DB stimulated during the task
- On each trial they chose between:
  - Low-risk option: 80% probability of winning €12
  - High-risk option: 40% probability of winning €18
- The low-risk option was the optimal strategy for maximizing cumulative reward.
- 150 trials divided into 3 blocks of 50 trials, breaks in between.



# TASK DETAILS



## **Trial Structure**

- 2 s baseline period
- Alternating presentation of the two symbols until a choice was made
- Feedback for chosen option
- Feedback for unchosen option
- Next trial begins

## **Recordings**

- EMG –Electrode placed on the neck
- LFP –Recorded from DBS electrodes implanted in the subthalamic nucleus
- (STN) within the basal ganglia



# FILES

## JSON

- Local Field Potential (LFP) data stored as per-recording dictionaries
- Sampling rate: 250 Hz
- Contains packet loss → needs correction and gap handling
- Must be synchronized with .OTB+ files for analysis alignment

## OTB+

- EMG recordings with two auxiliary channels: task\_events and response
- Task events encoded in trigger channel via amplitude (trial start, stim1, stim2, feedback1, feedback2)
- Sampling rate: 2000 Hz
- Task events and response times synchronized with EMG signal

## .LOG

- Text files describing trial-level events (symbol onset, risk level, feedback, choices)
- Include metadata: high/low risk symbols, feedback content, and patient decisions
- Original timestamps are unreliable
- All timing replaced with synchronized timestamps from .OTB+ files



## OPENING OTB FILES

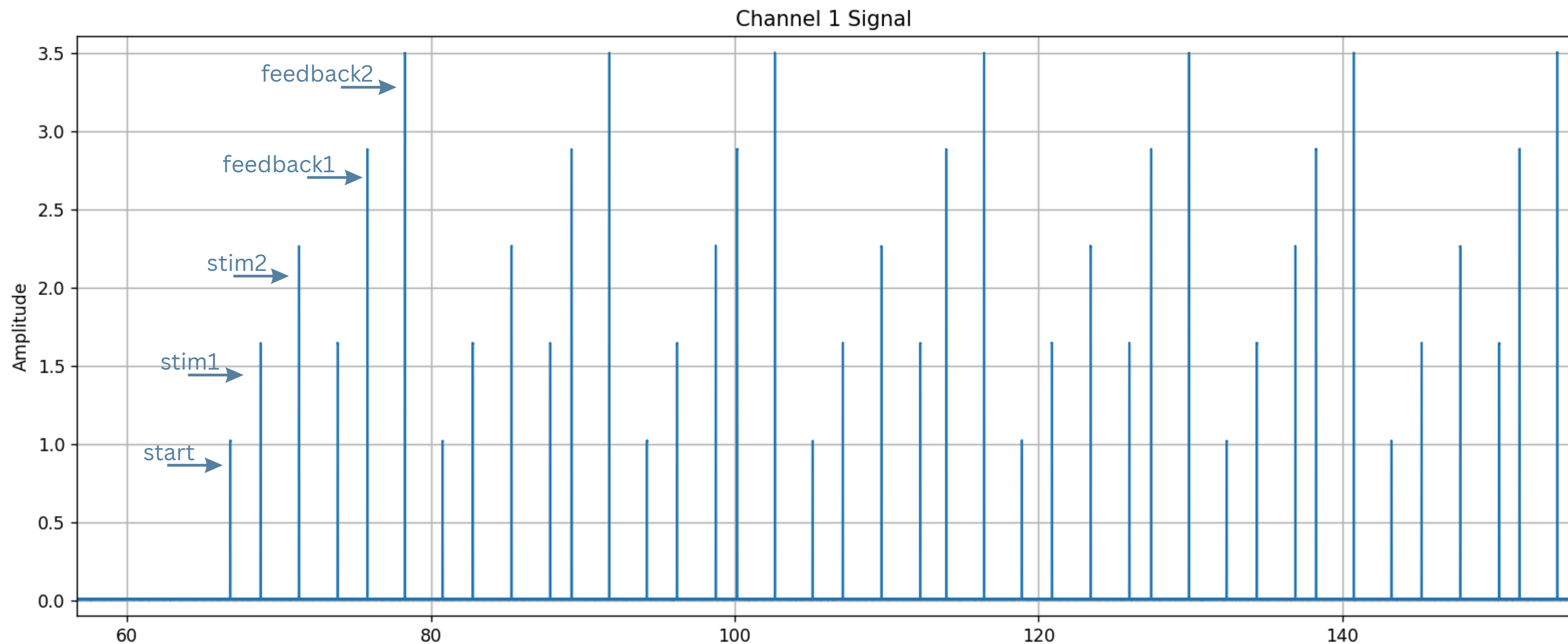
- OTB+ files were opened and saved using MATLAB.
- Packet loss was handled by the code
- Data were saved into two .mat files – one for each AUX
- .mat files were converted to .csv for easier processing in Python.

## PARSING LOGS

- Each .log file was parsed and important trial information was extracted into a single CSV file.
- The parser reads the file line by line, detects trial boundaries, and collects relevant information for each trial.

# VISUALIZATION

To check the correctness of imported csv files and to aid in further analysis visualization functions were written.



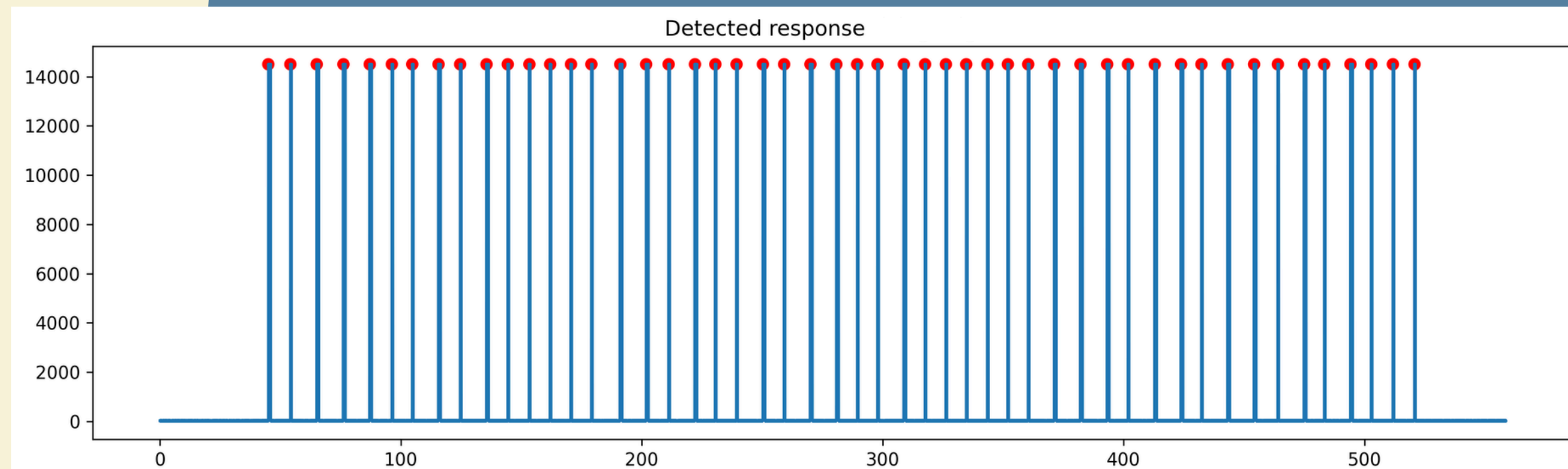
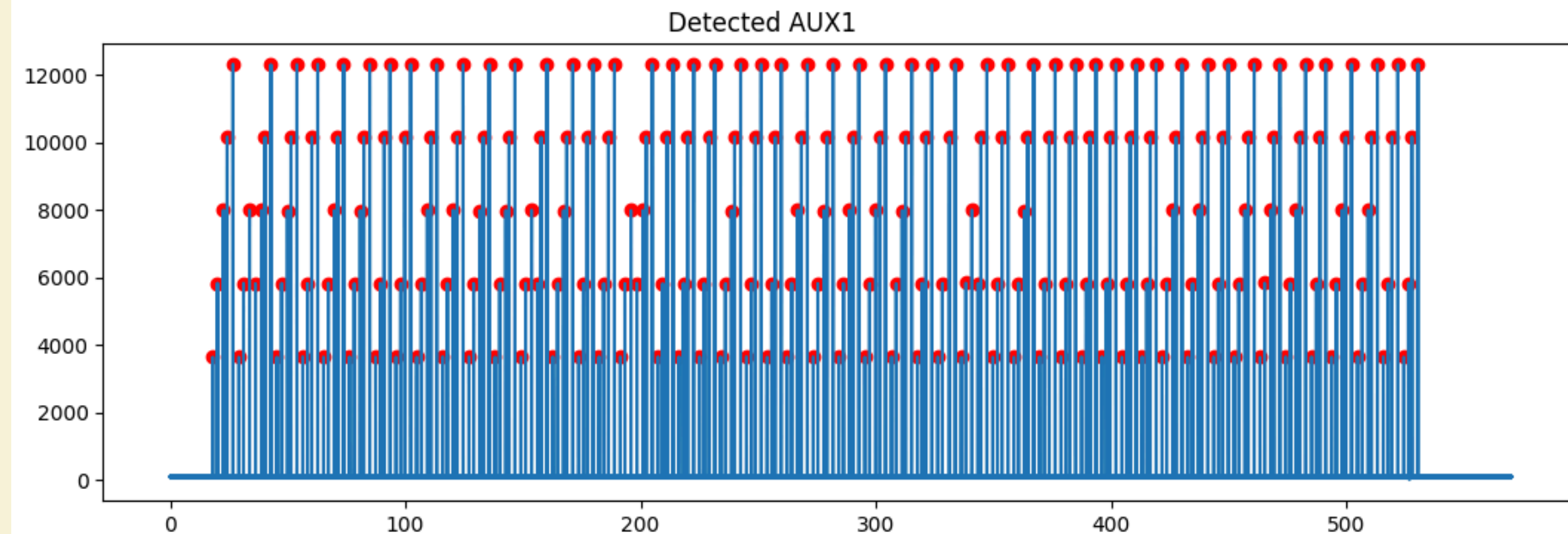
# EVENT TRIGGERS

- EMG signals were smoothed using a rolling median filter to reduce noise and suppress small local peaks.
- Events were detected using threshold-based band detection.
- Event timestamps were defined as the midpoint of each detected segment.
- A minimum gap between events was enforced to prevent duplicate detections.

## RESPONSE TRIGGER

- Response triggers were detected using a global amplitude threshold set at 50% of the maximum signal amplitude.
- Continuous segments above the threshold were identified as response events.
- The midpoint of each segment was used as the response timestamp (response\_ts).
- Reaction time was calculated as the difference between each response timestamp and the most recent valid preceding stimulus timestamp.

The validity was checked by comparing number of timestamps obtained from the signal with the number of corresponding events in the log file.

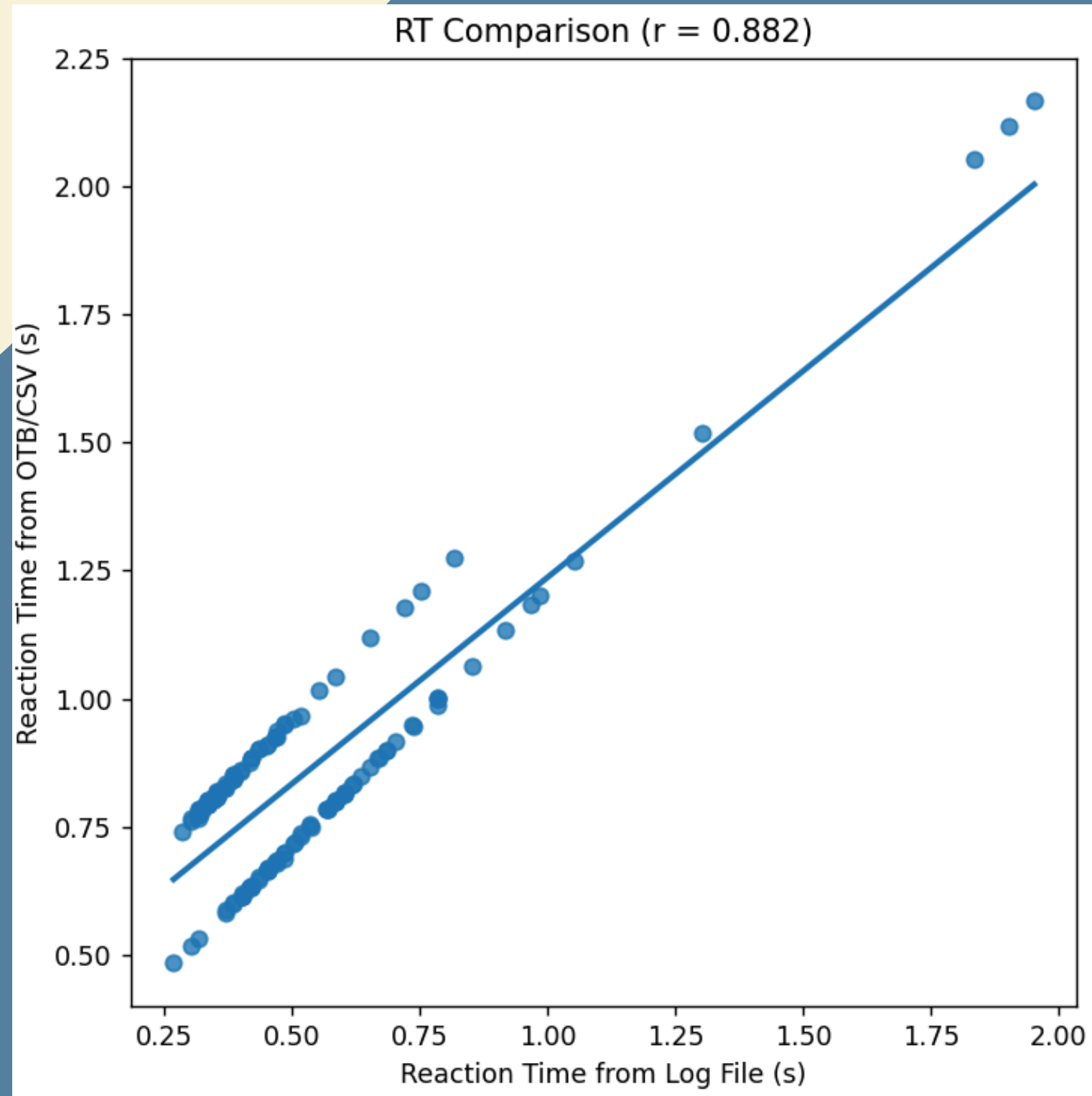




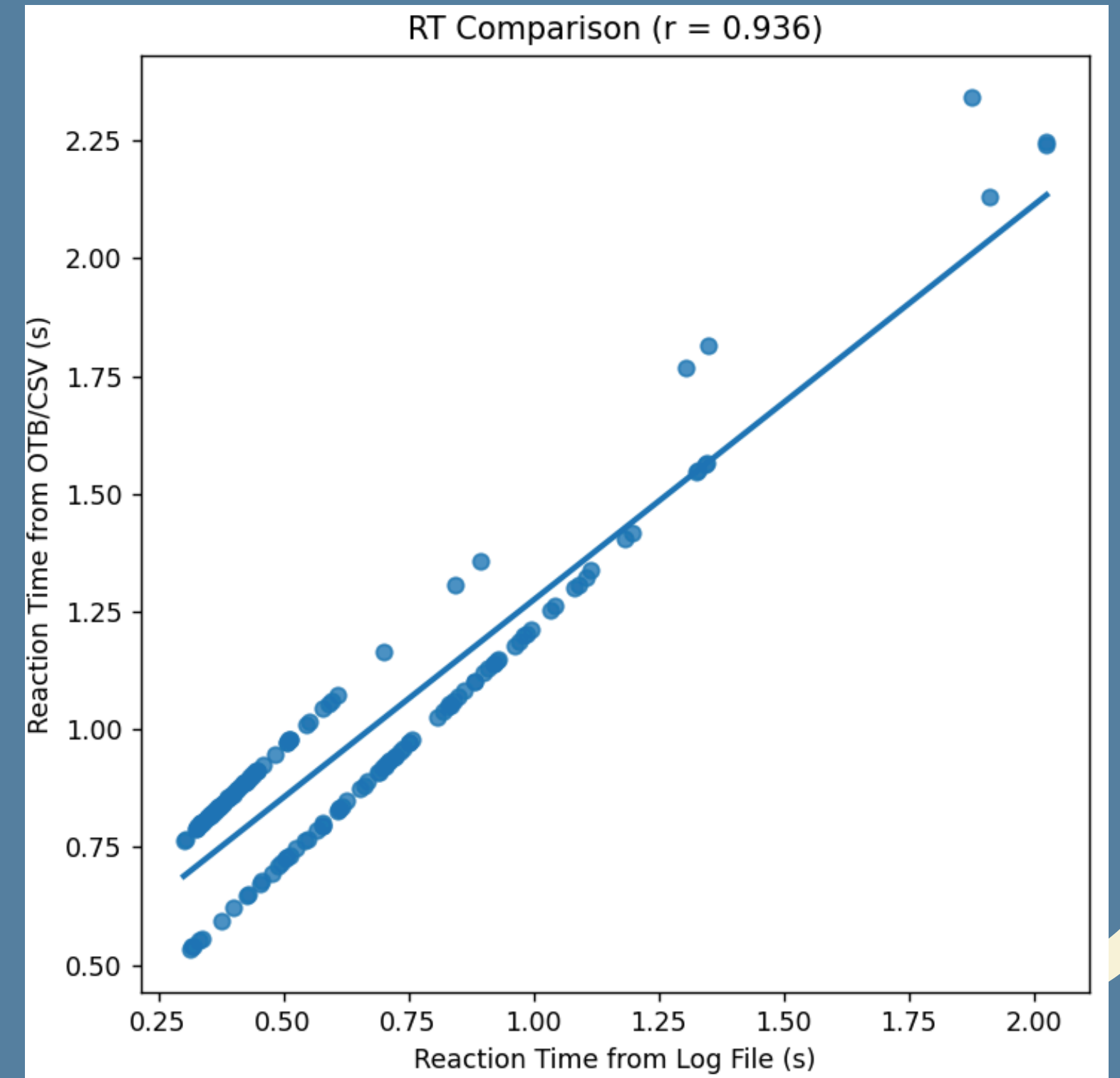
# CORRELATION LOG VS CSV

CORRELATION BETWEEN REACTION TIME COMPUTED FROM SIGNAL TIMESTAMPS AND RECORDED IN LOG FILES WAS CHECKED TO ENSURE VALIDITY. THE TWO SHOW STRONG POSITIVE CORRELATION, PROVING THE PIPELINE WORKED AS INTENDED.

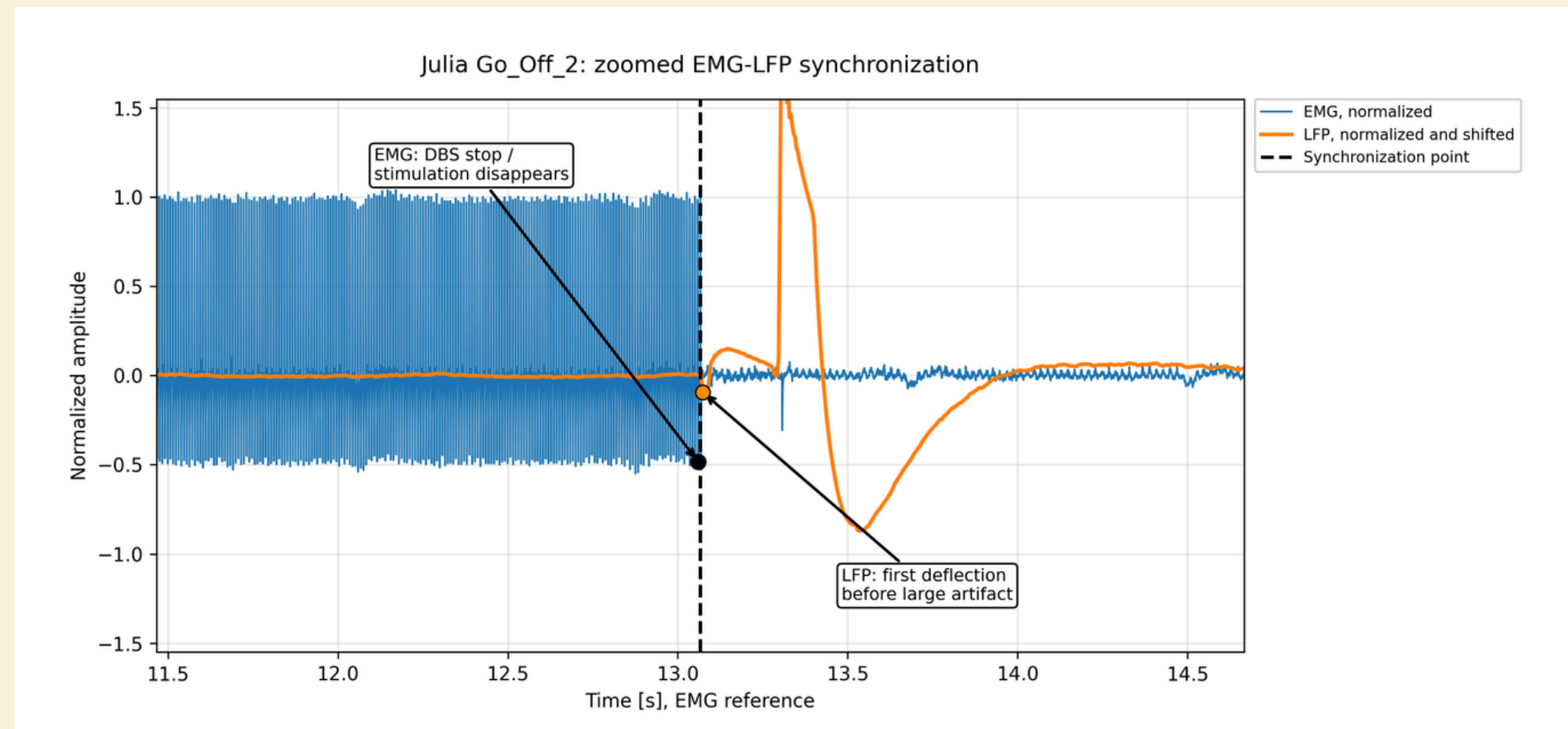
NATALIA



JULIA

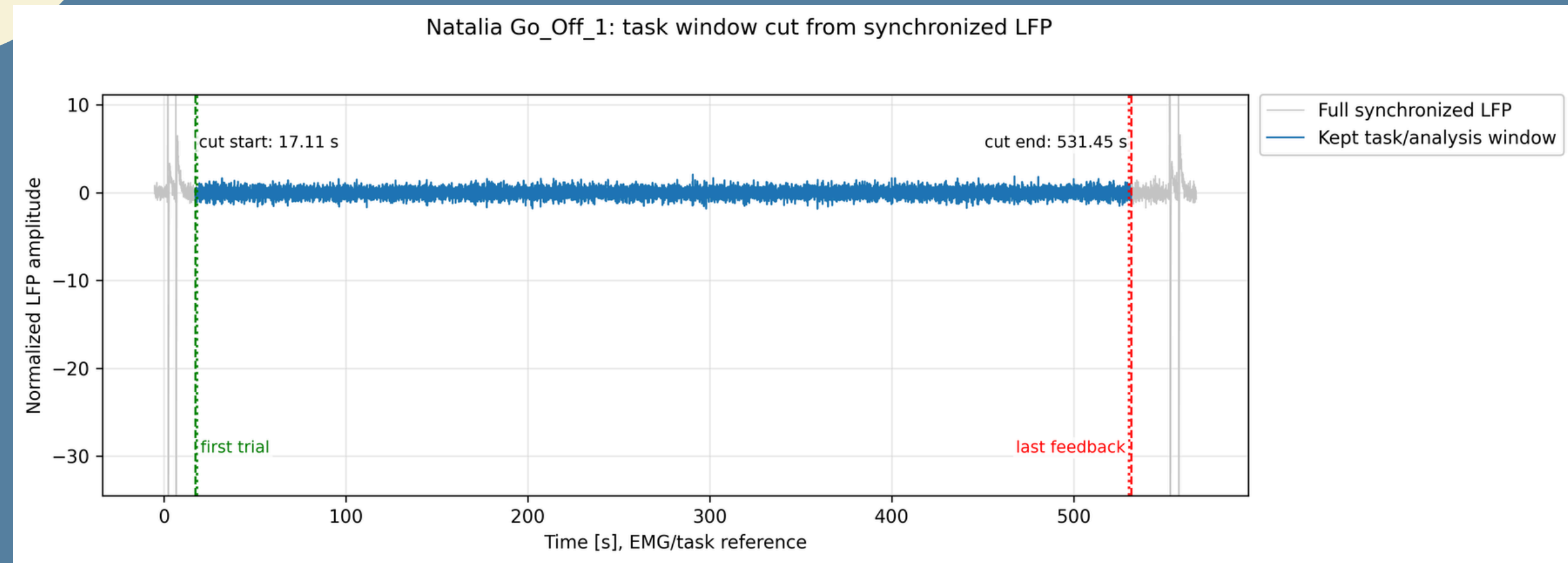


# EMG-LFP SYNCHRONIZATION

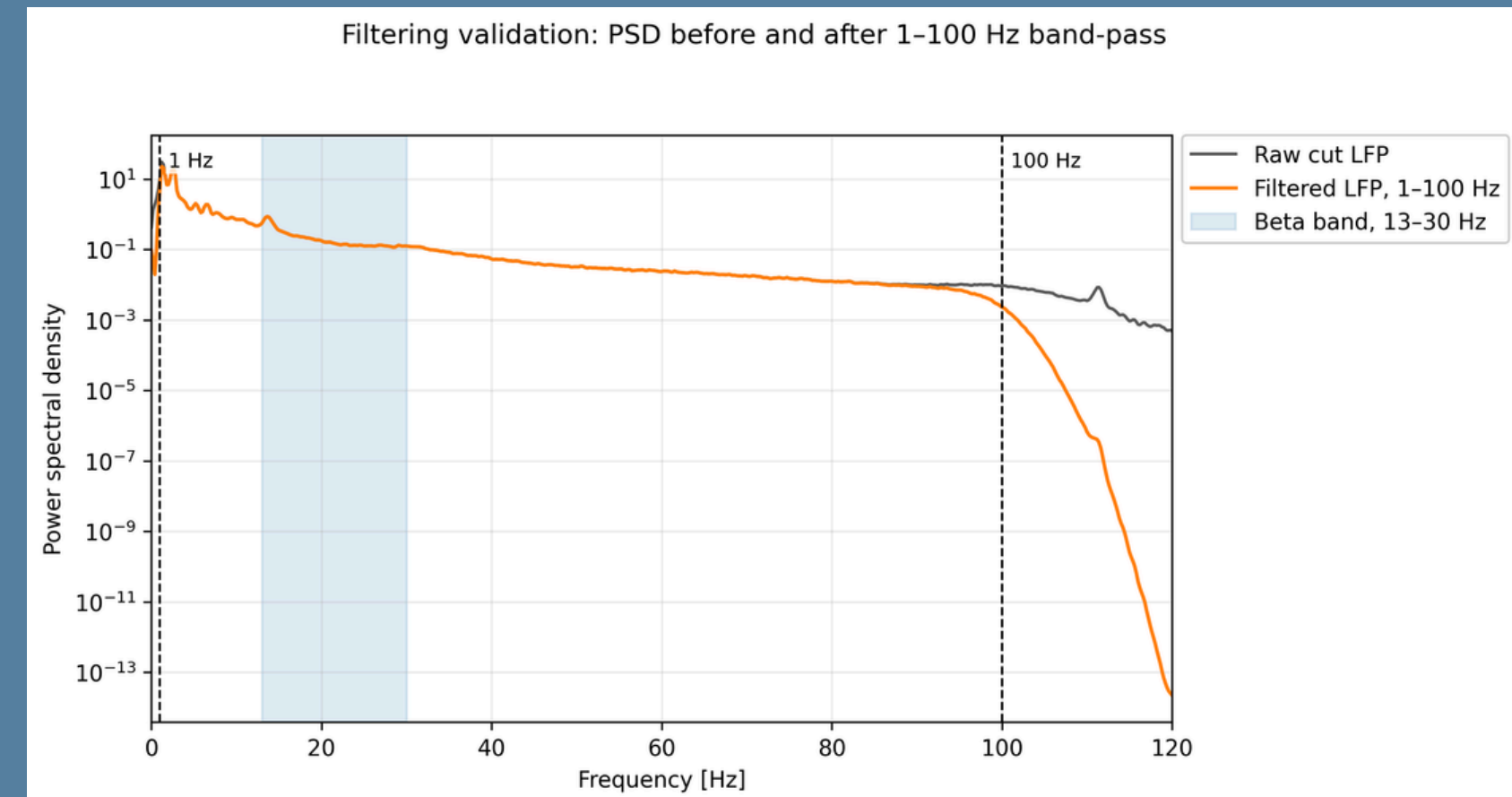


- Synchronization event: brief DBS interruption produced a visible artifact in both EMG and LFP signals.
- EMG reference point: selected at the last DBS-related oscillation before stimulation disappeared.
- LFP reference point: selected at the first clear deflection before the large DBS artifact.
- Time alignment: the LFP time axis was shifted so that the LFP synchronization point matched the EMG synchronization point.

# FILTERING AND PSD

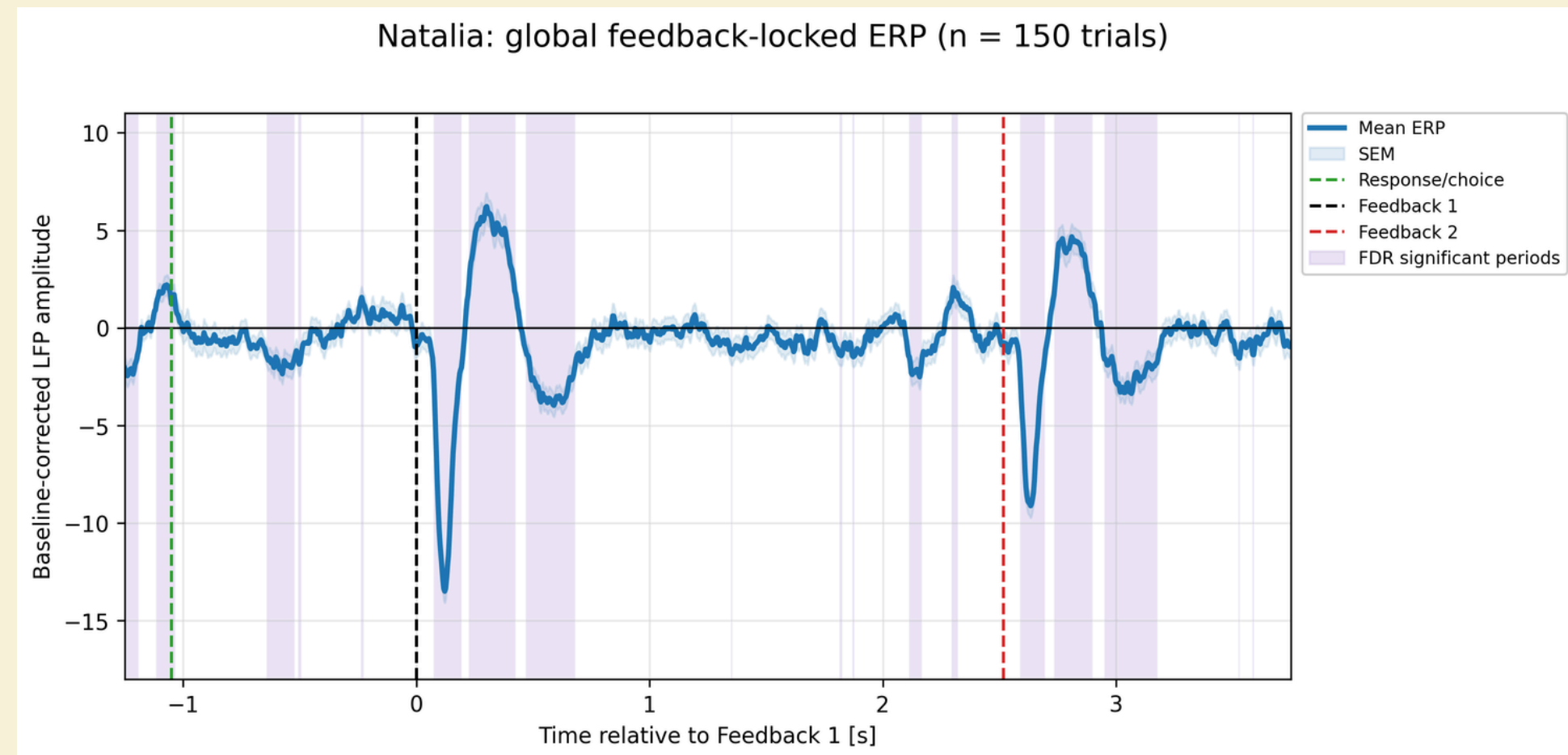


- Synchronization artifacts at the beginning and end of each block were excluded before filtering.
- The kept analysis window was defined using task timestamps: from first trial start – 1 s to last feedback 2 + 1 s.
- A 4th-order Butterworth band-pass filter, 1–100 Hz, was applied only to the cut task window.
- PSD validation confirmed that the filter preserved the relevant low-frequency activity, including the beta range, while attenuating activity outside the analysis range.



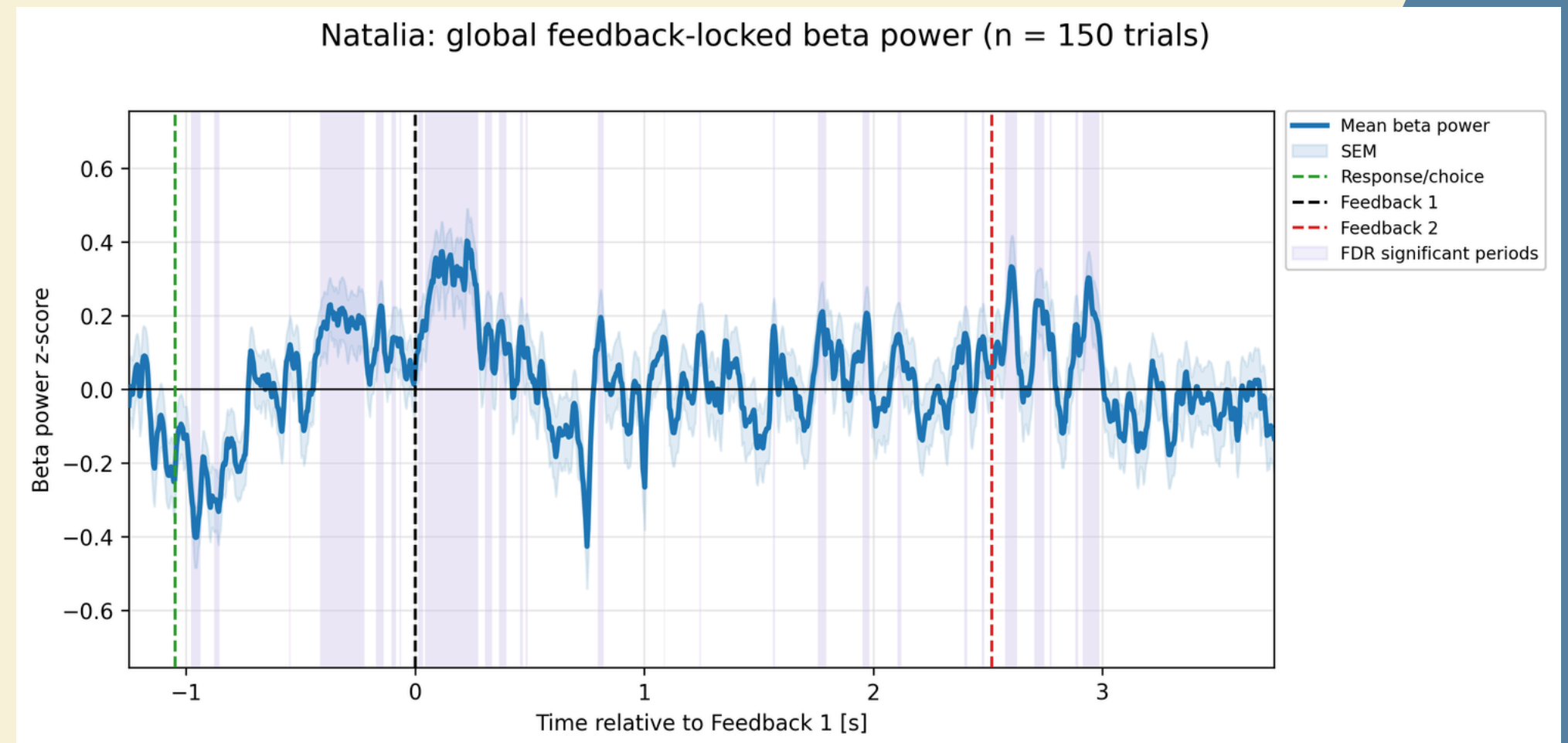
# EVENT RELATED POTENTIAL ANALYSIS

- ERP was calculated by averaging LFP epochs aligned to Feedback 1.
- Time 0 corresponds to Feedback 1.
- Choice and Feedback 2 were also marked to show the full task structure.
- The blue curve shows the mean baseline-corrected LFP response.
- The shaded area represents SEM across trials.
- Significant time periods were assessed using point-wise tests against zero with FDR correction.



# BETA-BAND ANALYSIS

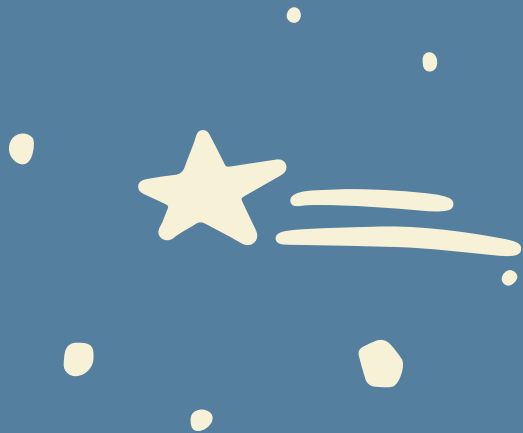
- LFP signals were filtered in the beta frequency range: 13–30 Hz.
- The Hilbert transform was applied to estimate instantaneous beta power.
- Beta power was normalized using a z-score baseline correction.
- Trials were aligned to Feedback 1, which was set as time 0.
- Response/choice and Feedback 2 were also marked to show the full task structure.
- Significant time periods were tested against zero and corrected using FDR.







# ANALYSIS

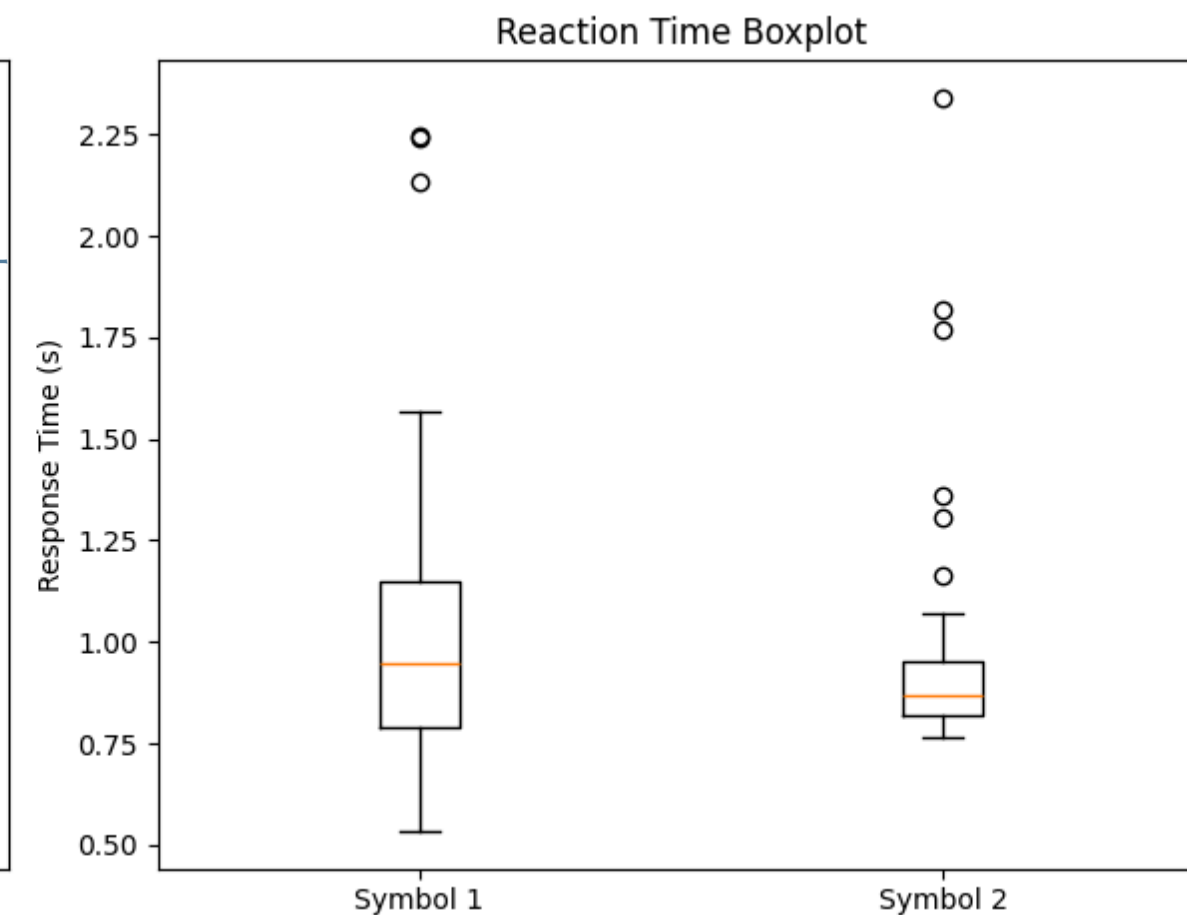
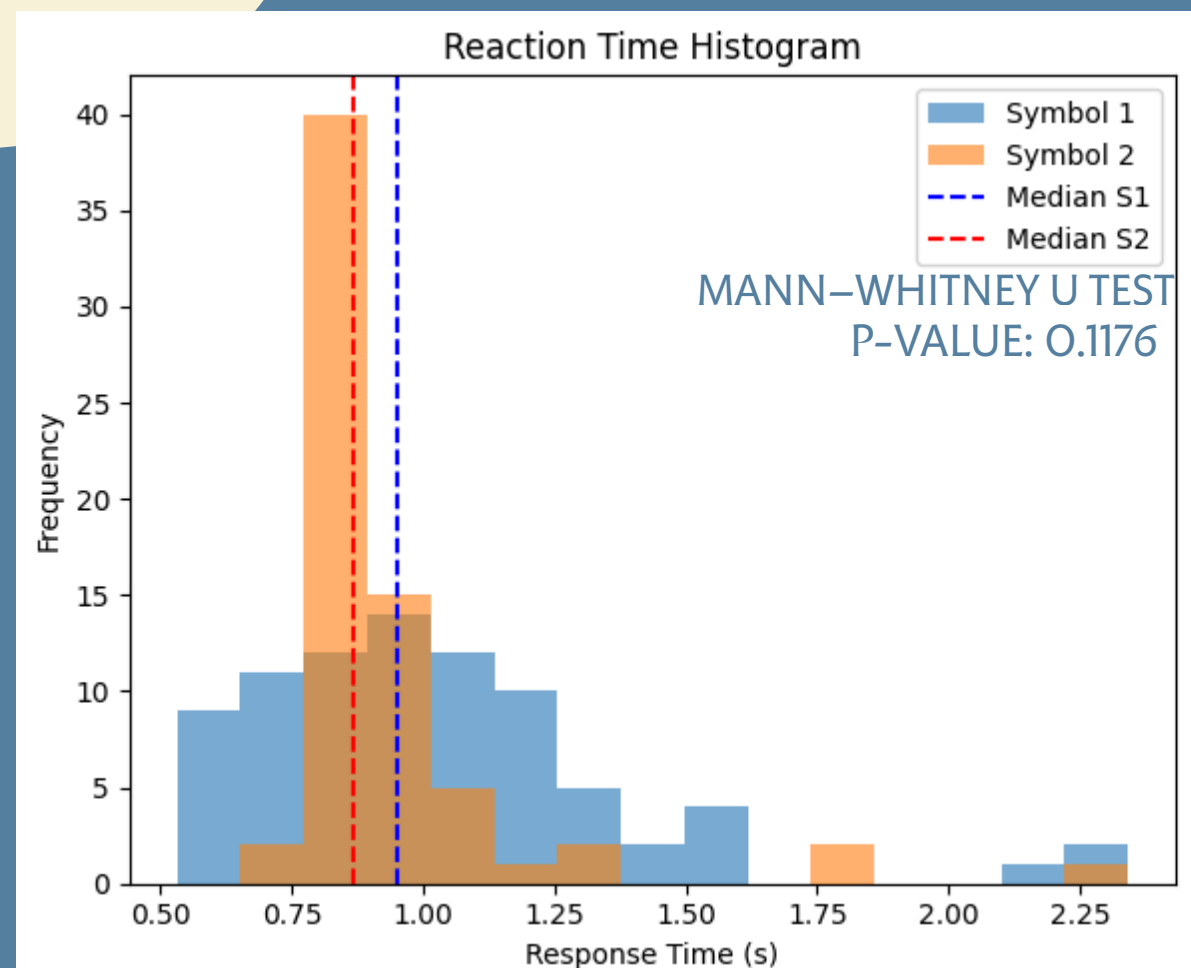


For simplicity, we assigned our own names to the patients we received, respectively.

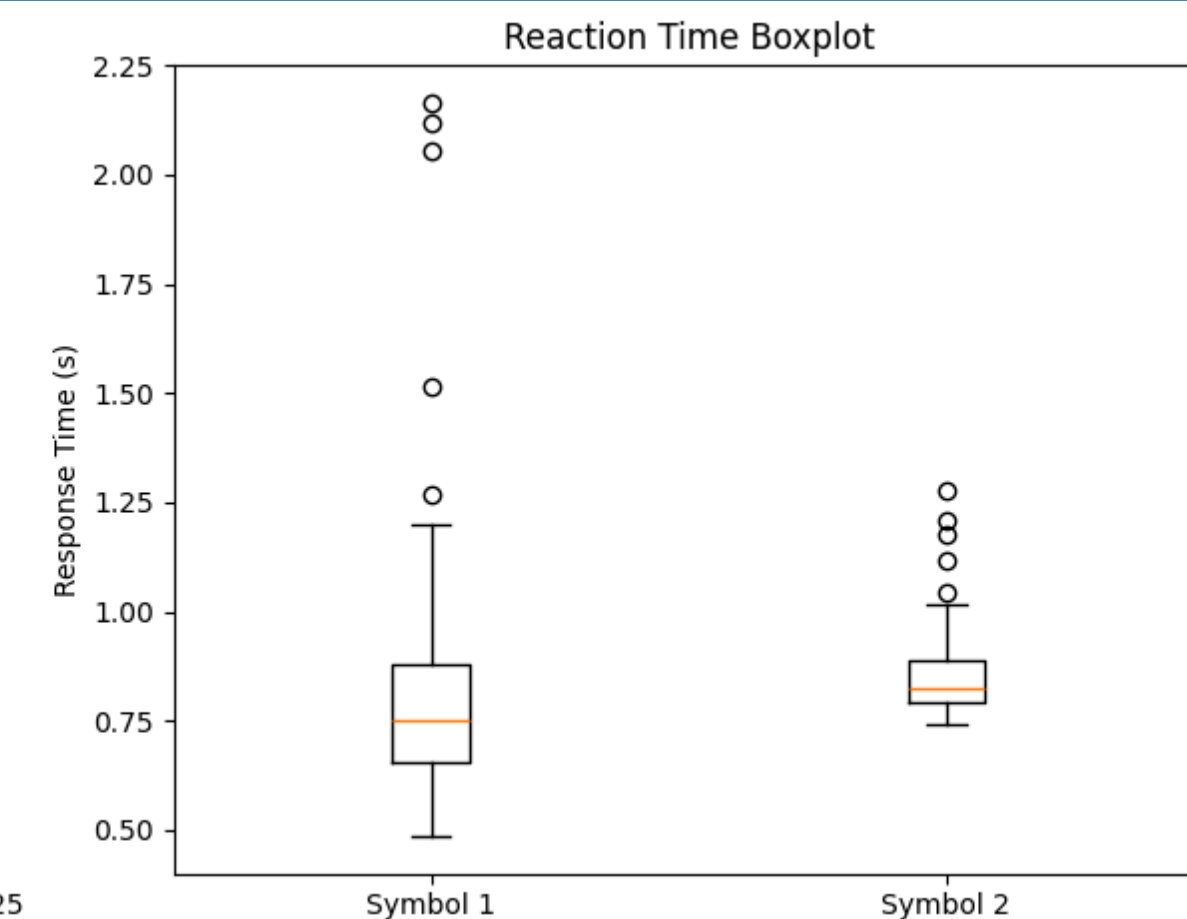
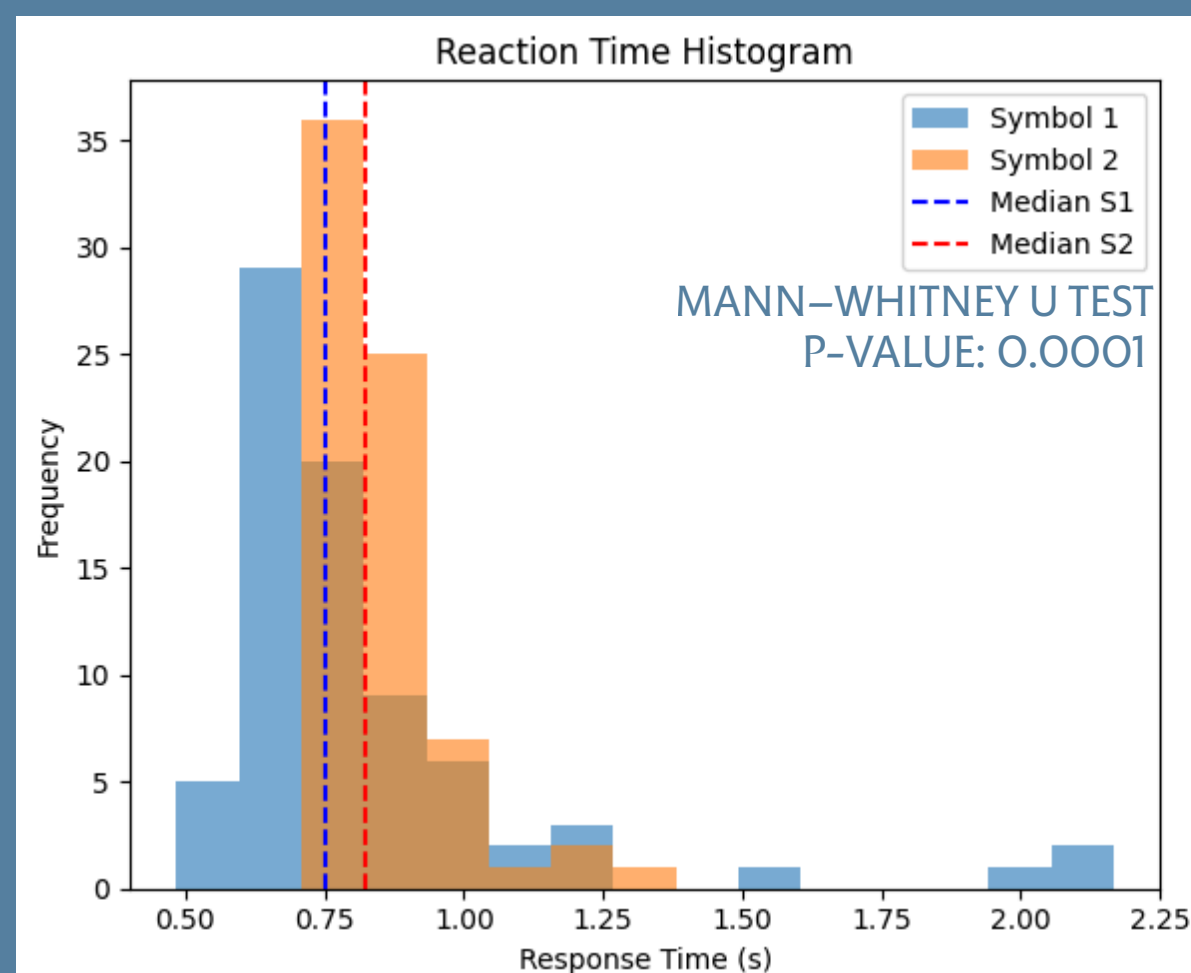


# REACTION TIME COMPARISON - STIM 1 / STIM 2

JULIA  
MEDIAN REACTION  
TIMES:  
CHOSEN SYMBOL 1: 0.9490 S  
CHOSEN SYMBOL 2: 0.8675 S

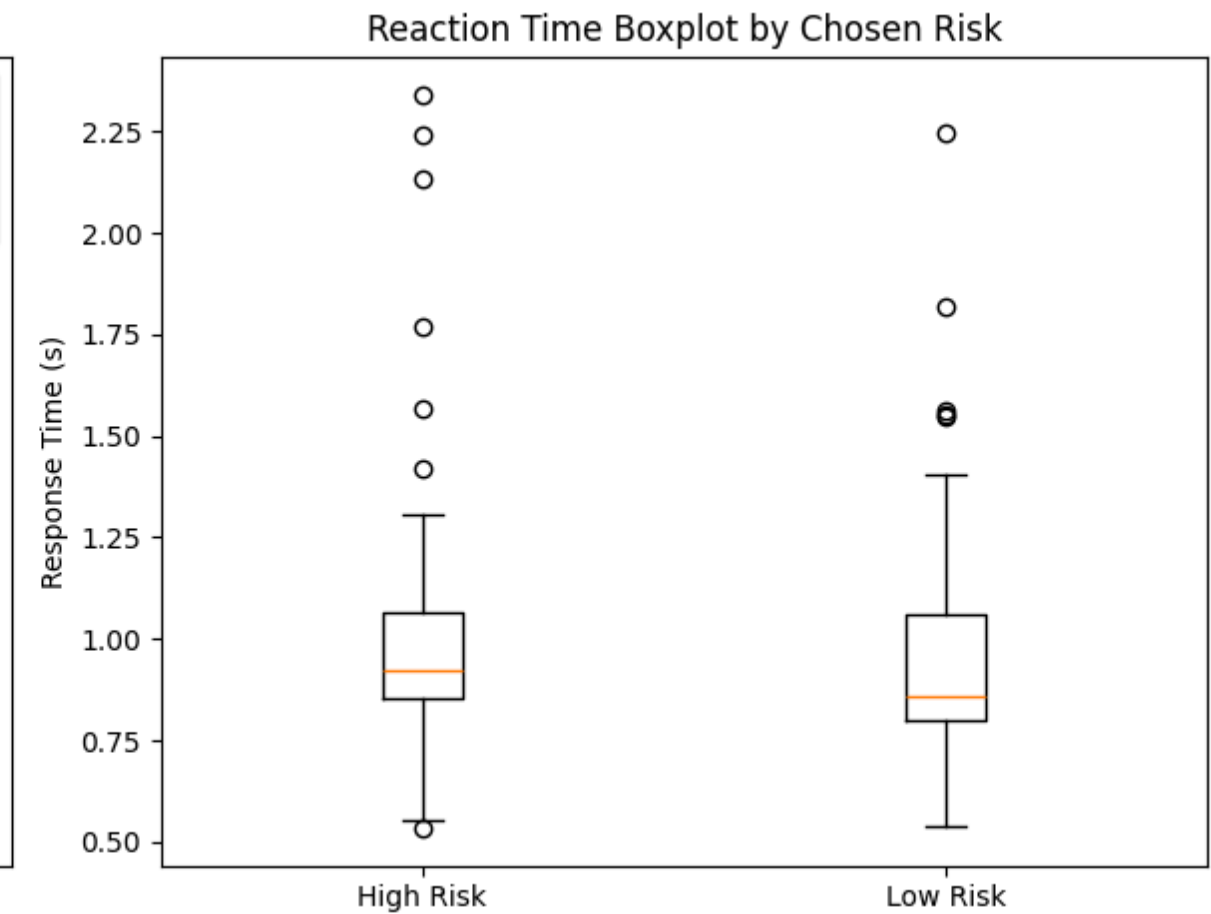
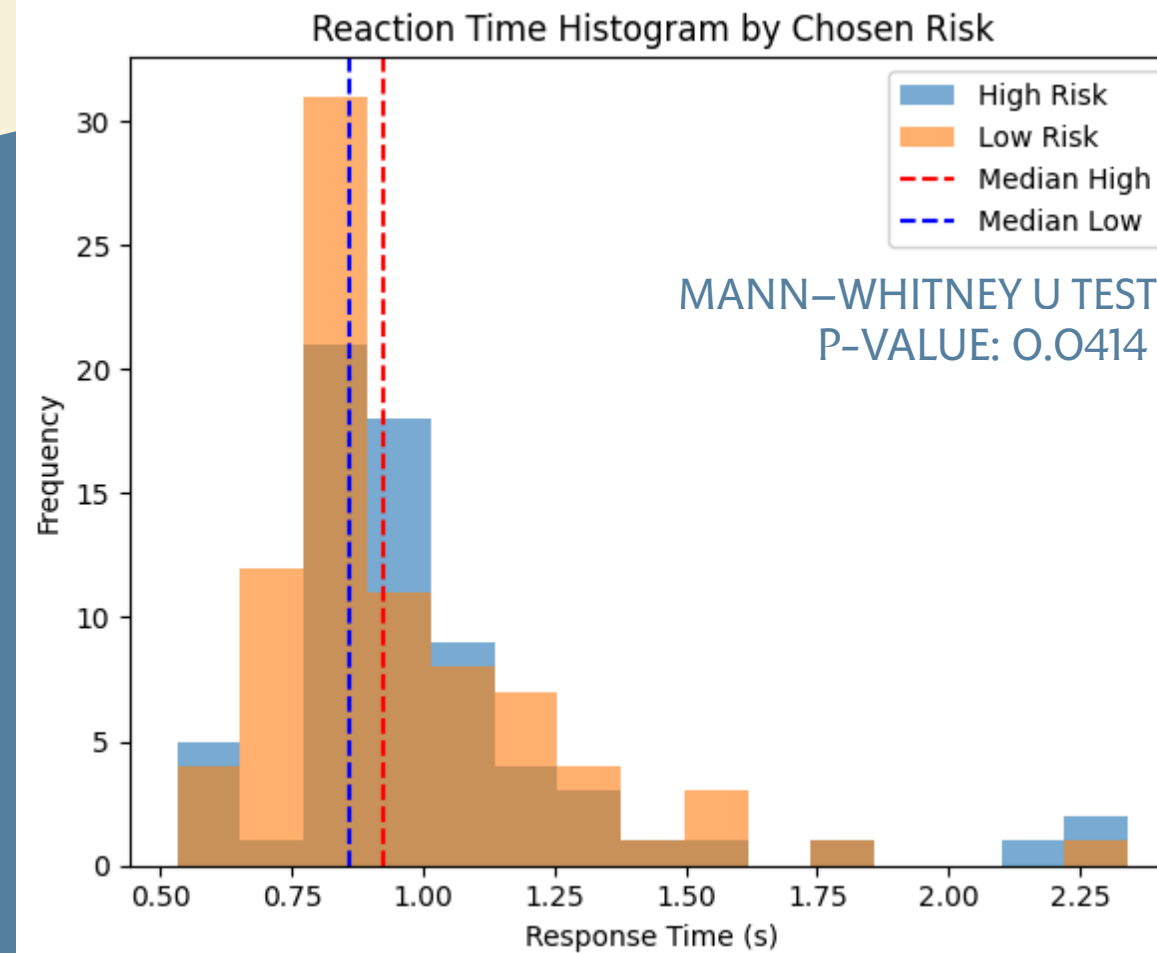


NATALIA  
MEDIAN REACTION TIMES:  
CHOSEN SYMBOL 1: 0.7515  
CHOSEN SYMBOL 2: 0.8225 S

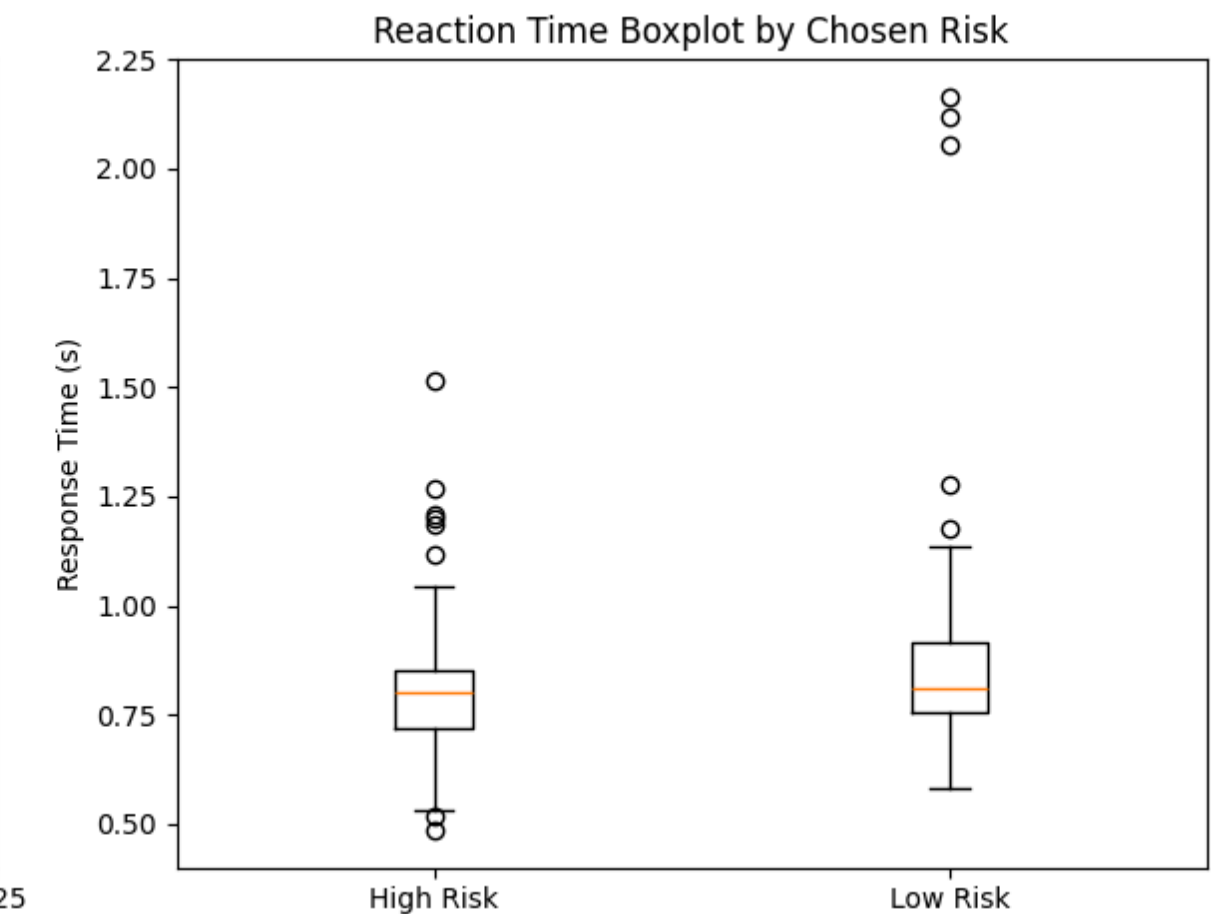
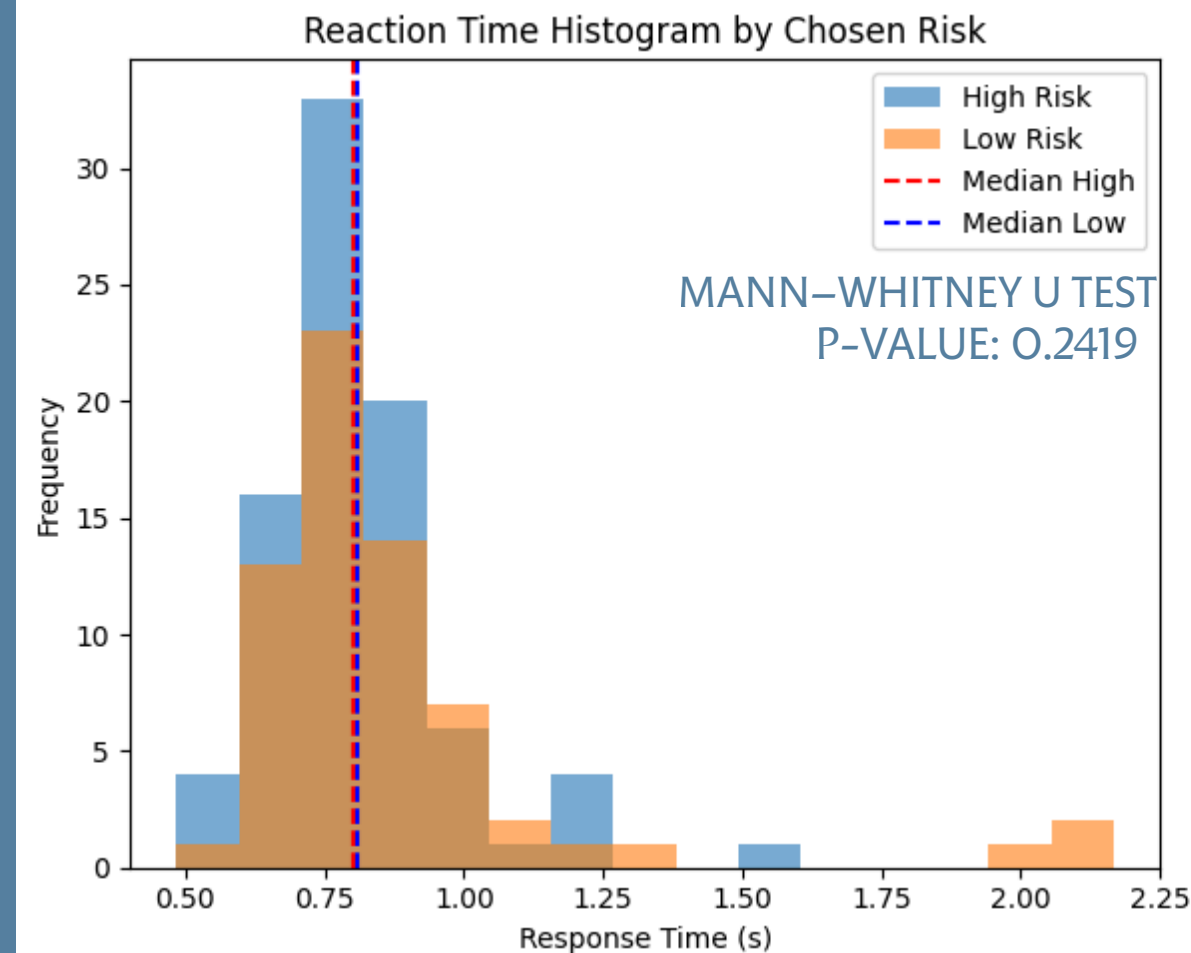


# REACTION TIME COMPARISON - LOW / HIGH RISK

JULIA  
MEDIAN REACTION TIMES:  
HIGH RISK CHOICE: 0.9240 S  
LOW RISK CHOICE : 0.8590 S



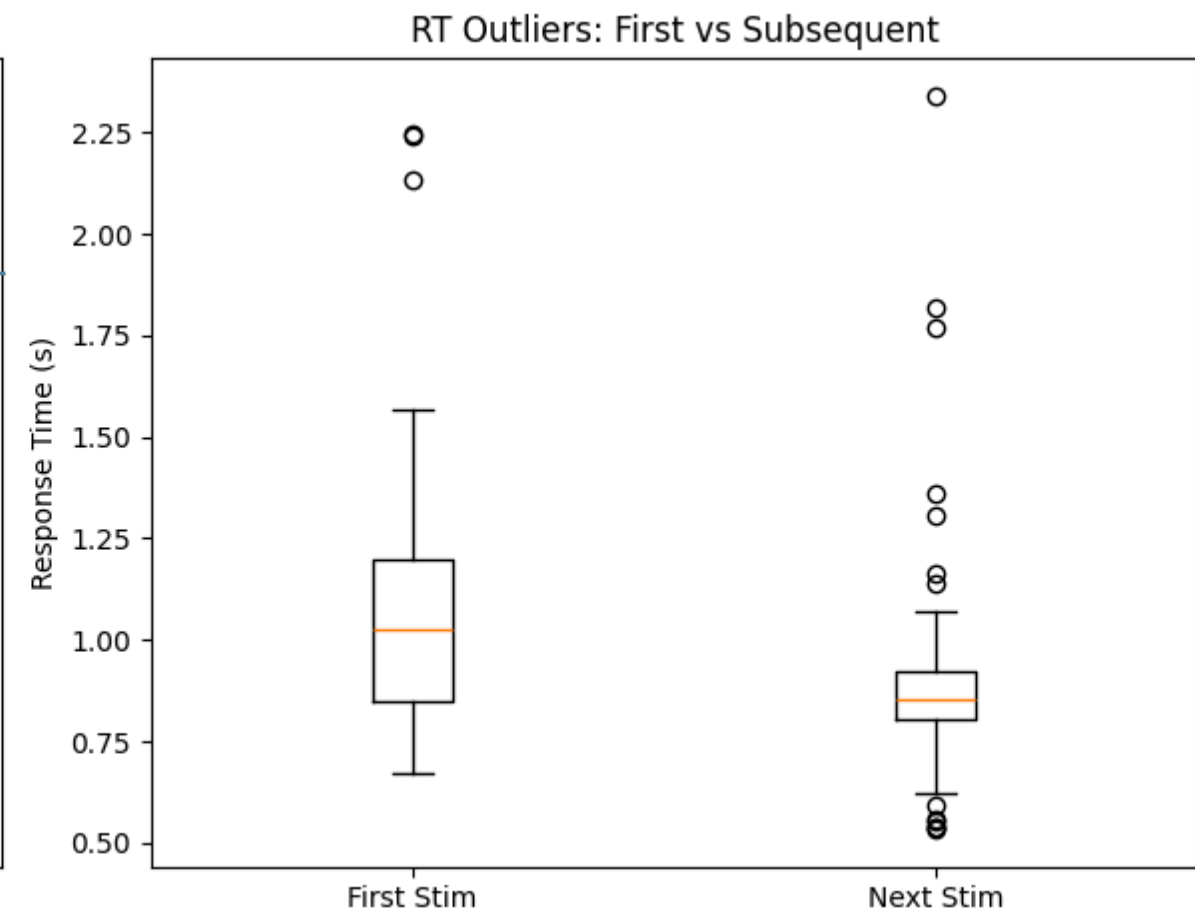
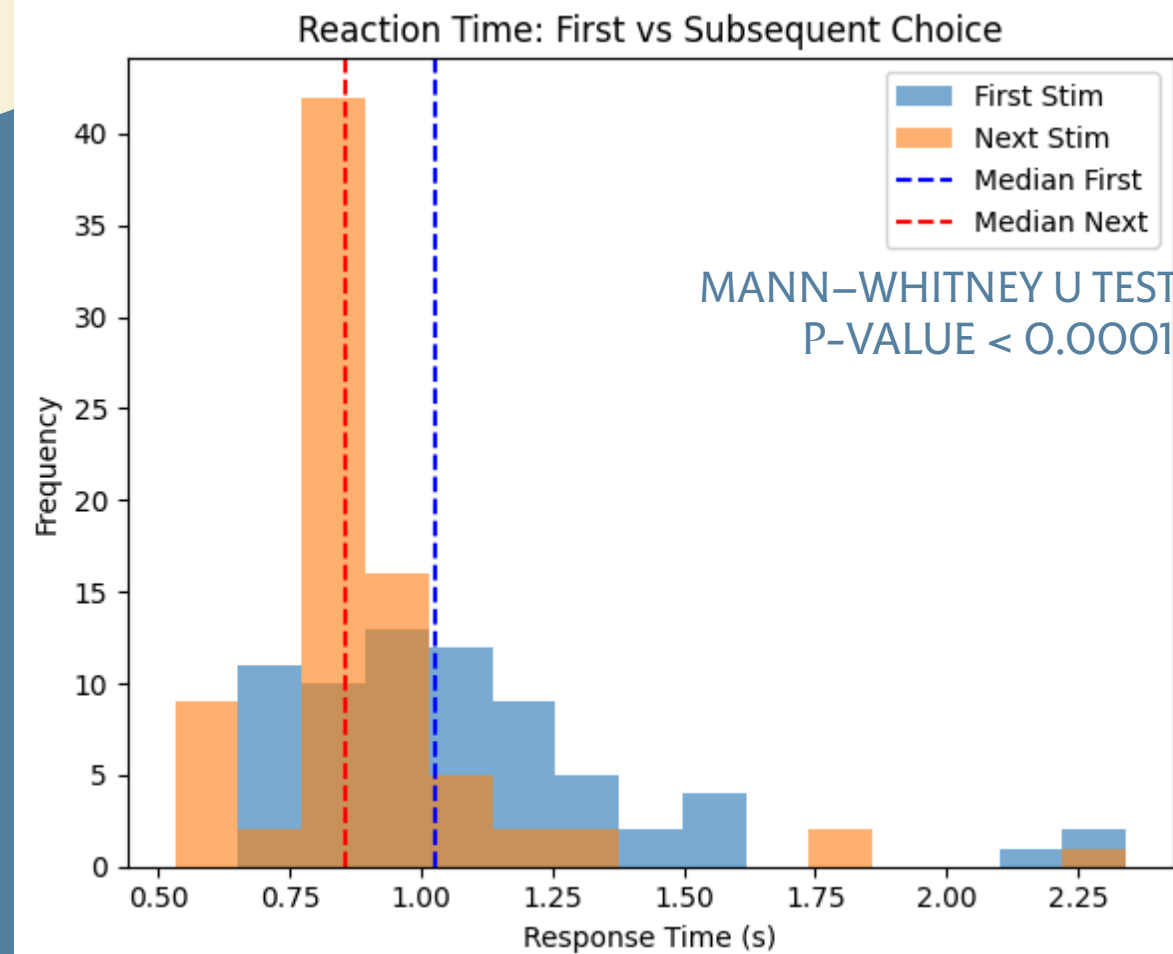
NATALIA  
MEDIAN REACTION TIMES:  
HIGH RISK CHOICE: 0.8000 S  
LOW RISK CHOICE : 0.8090 S



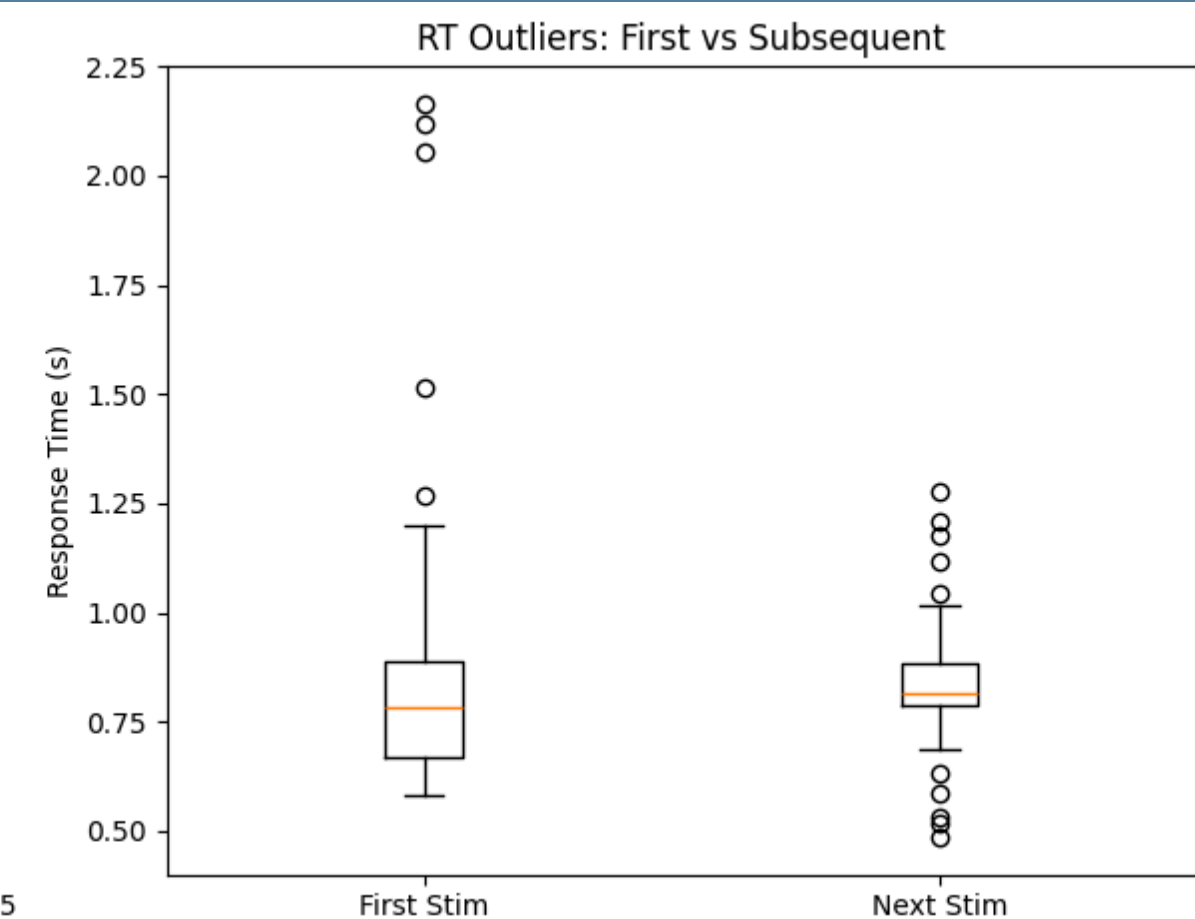
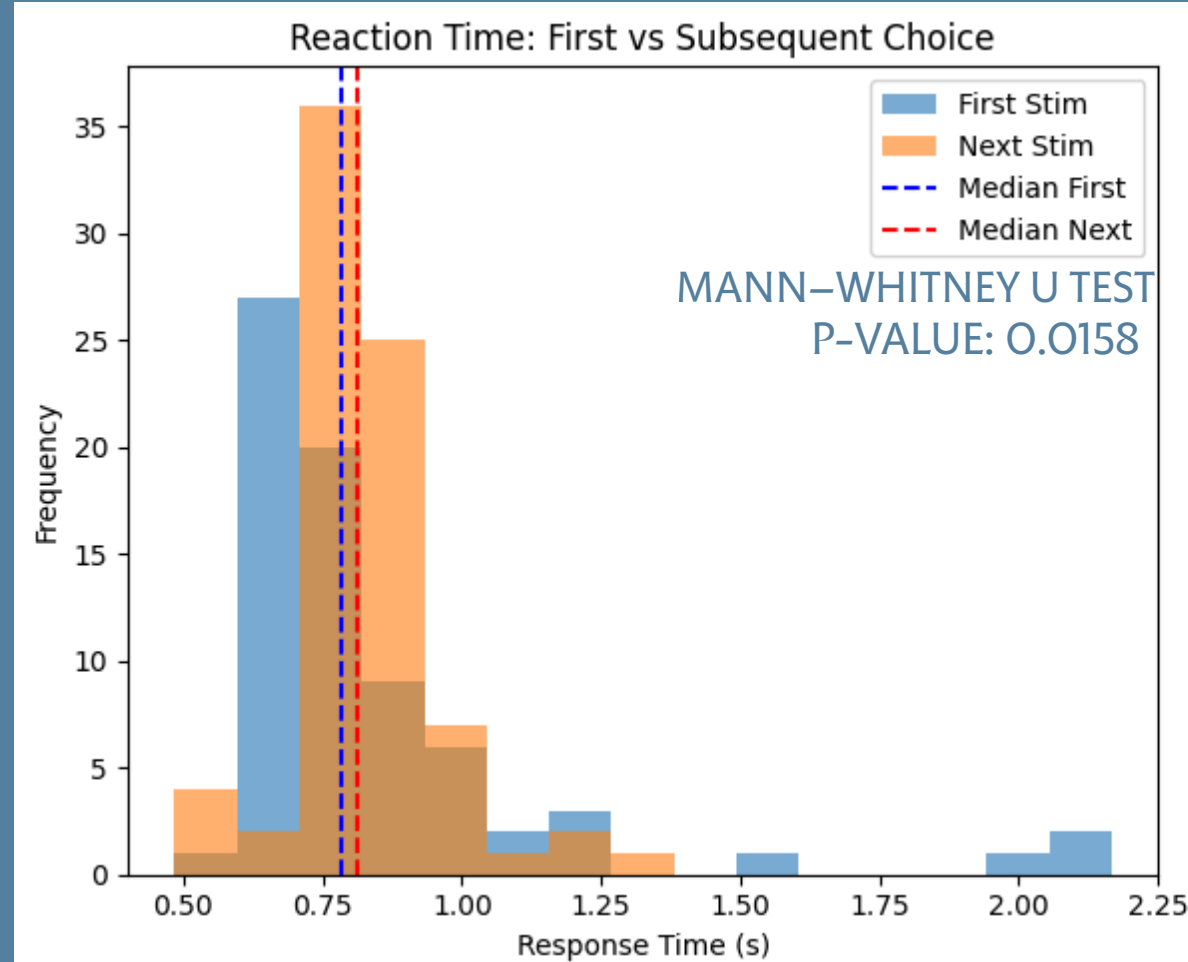


# REACTION TIME COMPARISON – FIRST STIM SEEN / SUBSEQUENT STIMS

JULIA  
MEDIAN REACTION TIMES:  
FIRST STIM CHOICE : 1.0270 S  
SUBSEQUENT STIM CHOICE : 0.8550 S



NATALIA  
MEDIAN REACTION TIMES:  
FIRST STIM CHOICE : 0.7845 S  
SUBSEQUENT STIM CHOICE : 0.8135 S

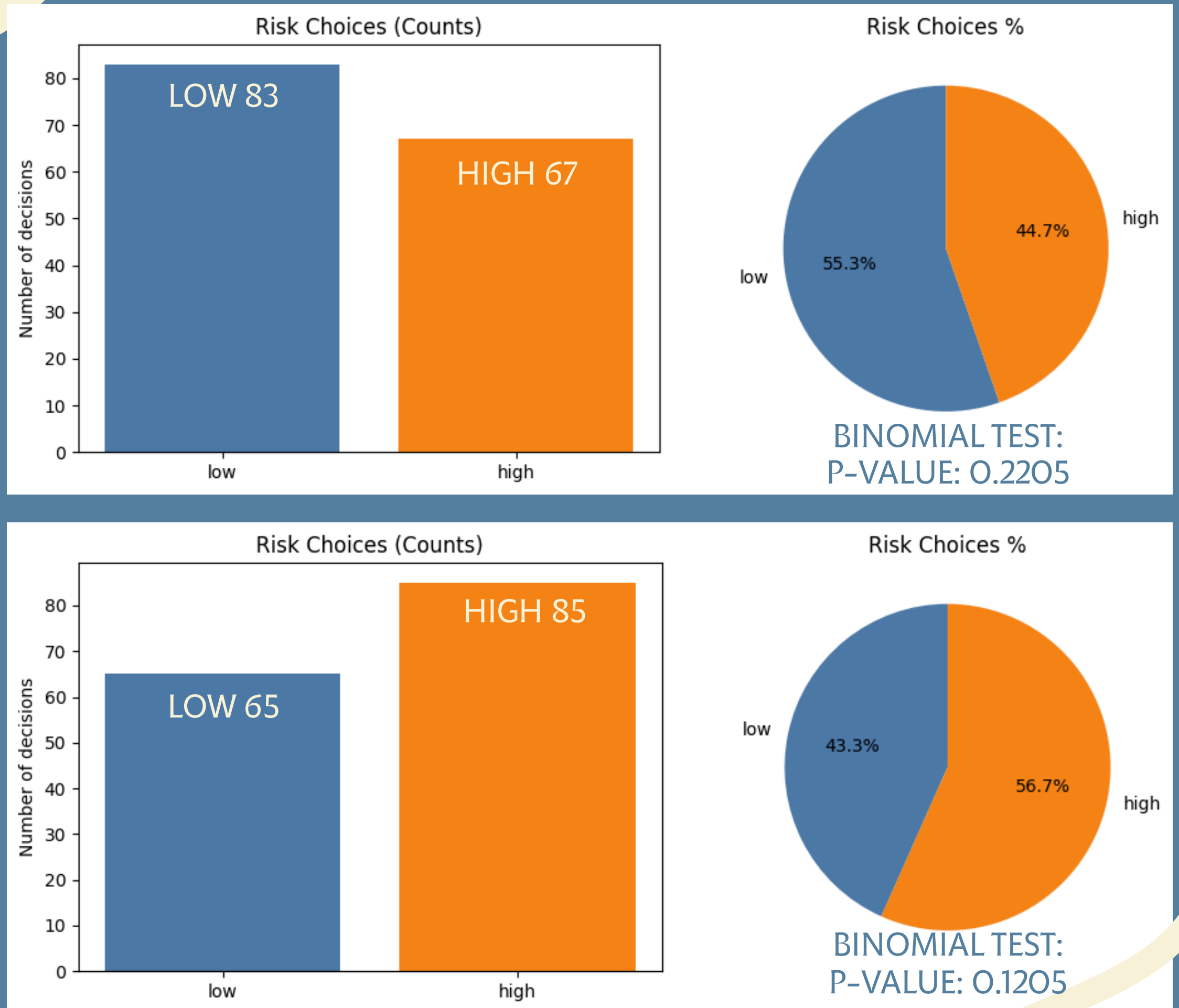


# NUMBER OF HIGH / LOW RISK CHOICES

JULIA

**CHI2 TEST HO:** THERE IS NO  
DIFFERENCE BETWEEN  
PATIENTS  
CHI2: 3.854  
P-VALUE: 0.0496

NATALIA



# CHOICE CHANGE OVER TIME

JULIA

BLOCK 1 VS BLOCK 2

$\text{CHI}^2 = 4.040$

P-VALUE = 0.0444

HIGH-RISK CHOICE RATE: 56.00% VS 34.00%

BLOCK 2 VS BLOCK 3

$\text{CHI}^2 = 0.673$

P-VALUE = 0.4122

HIGH-RISK CHOICE RATE: 34.00% VS 44.00%

NATALIA

BLOCK 1 VS BLOCK 2

$\text{CHI}^2 = 0.640$

P-VALUE = 0.4236

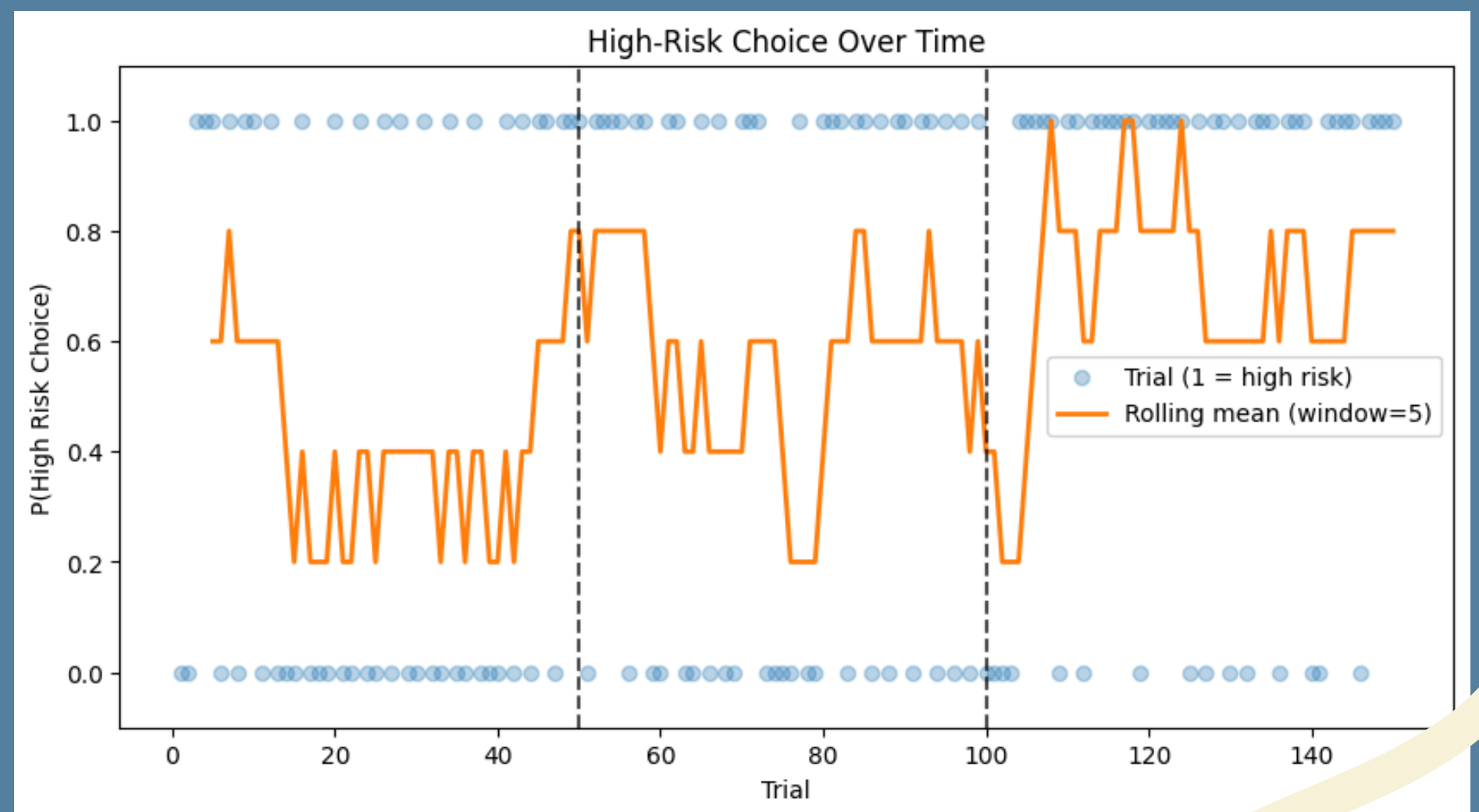
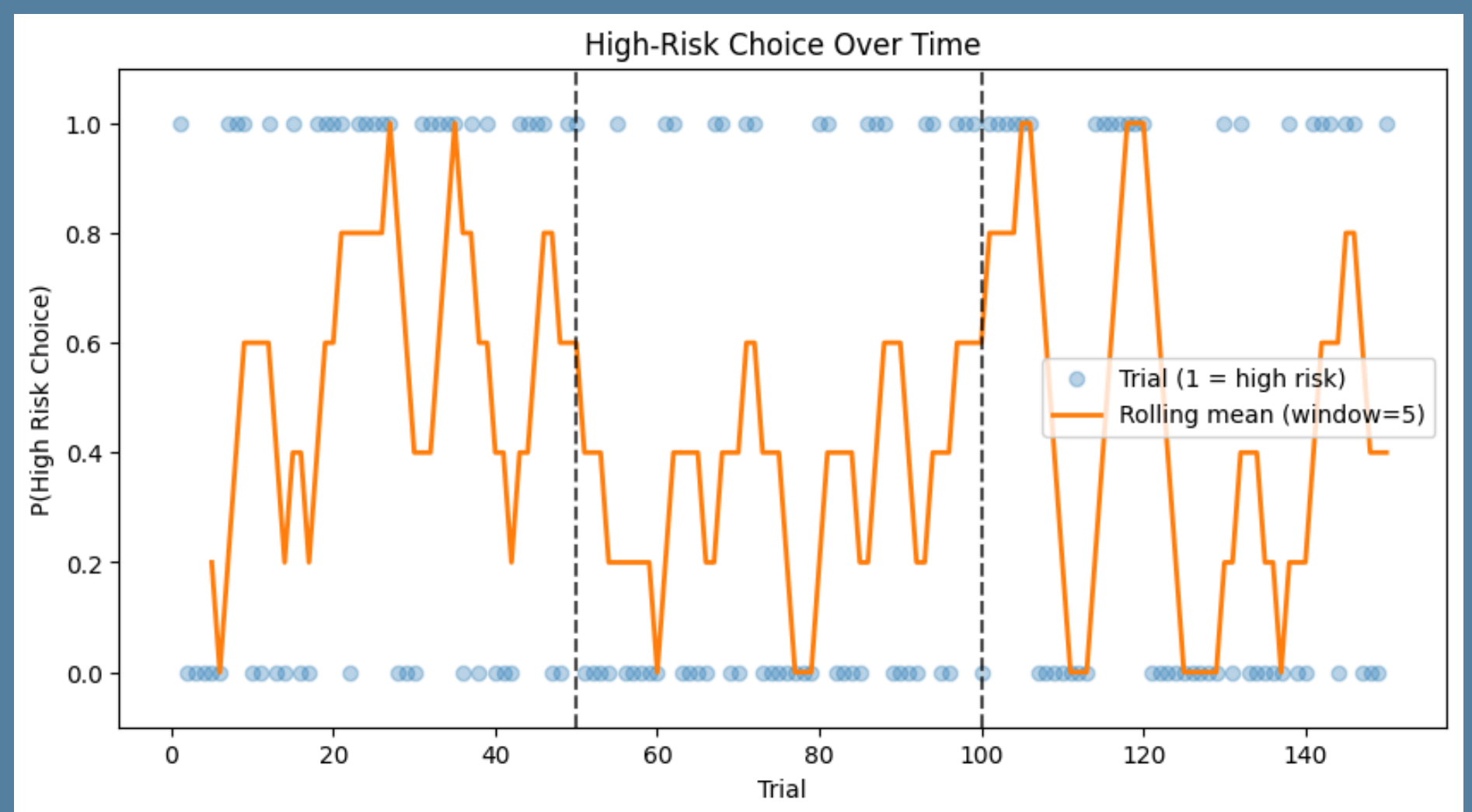
HIGH-RISK CHOICE RATE: 44.00% VS 54.00%

BLOCK 2 VS BLOCK 3

$\text{CHI}^2 = 2.746$

P-VALUE = 0.0975

HIGH-RISK CHOICE RATE: 54.00% VS 72.00%



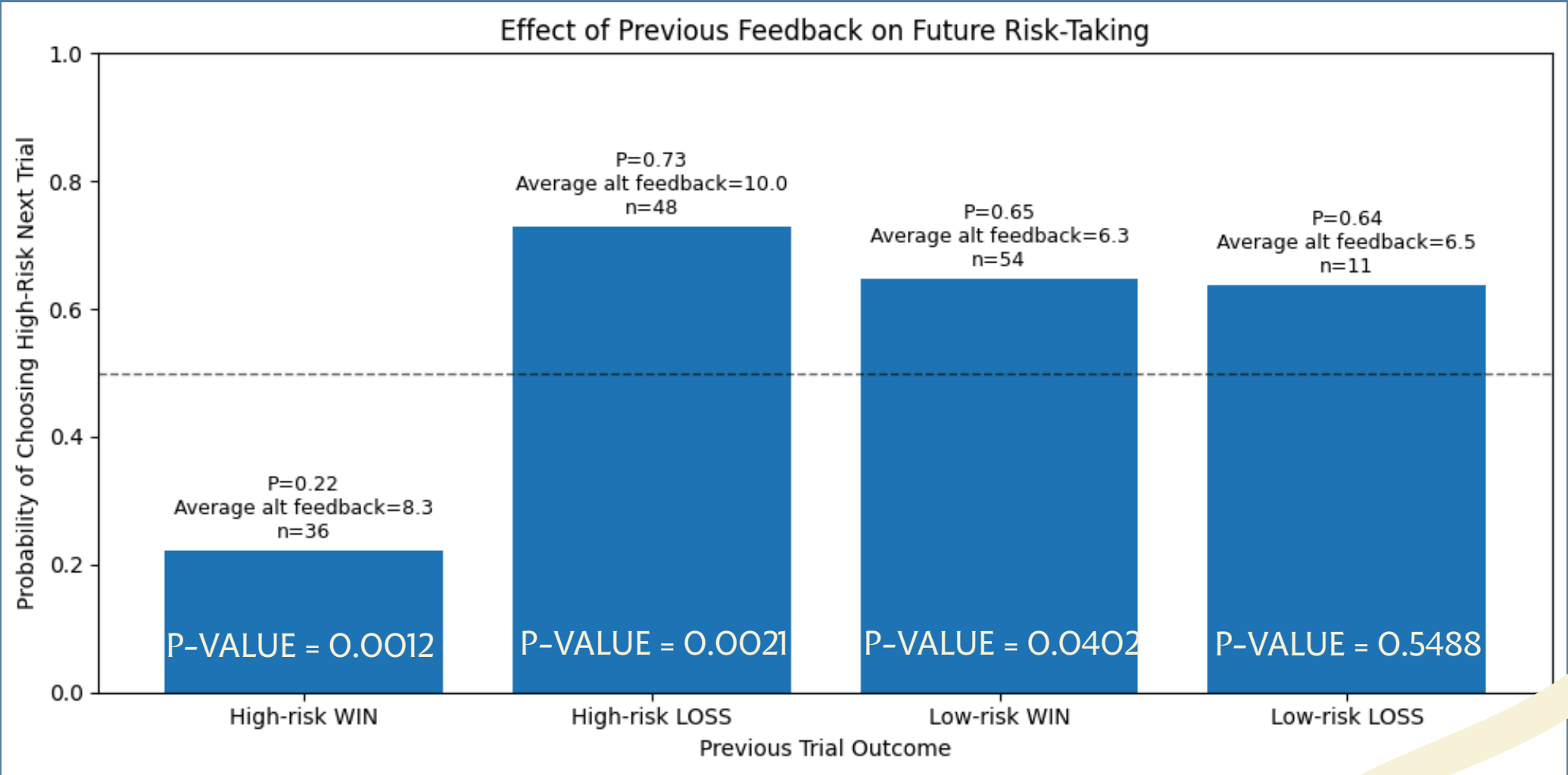
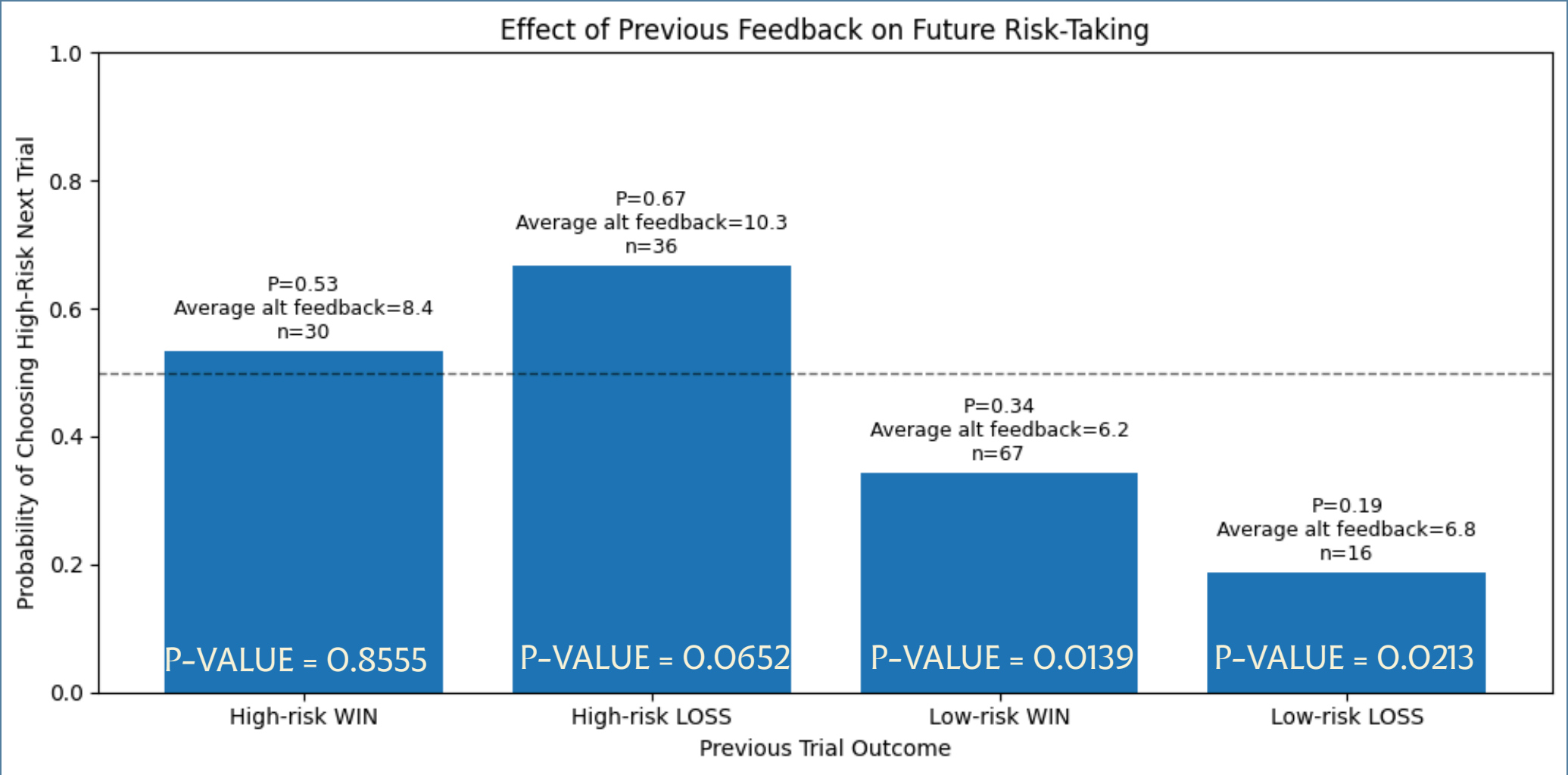
# EFFECT OF PREVIOUS FEEDBACK

JULIA  
 $\text{CHI}^2 = 15.224$   
P-VALUE = 0.0016

TWO TESTS PERFORMED:  
**CHI2 HO:** PREVIOUS FEEDBACK DOES NOT INFLUENCE NEXT DECISION  
**BINOMIAL TEST HO:** THE PROBABILITY OF CHOOSING HIGH RISK IN EACH CONDITION IS RANDOM

BINOMIAL TEST VALUES PRESENTED DIRECTLY ON THE GRAPH

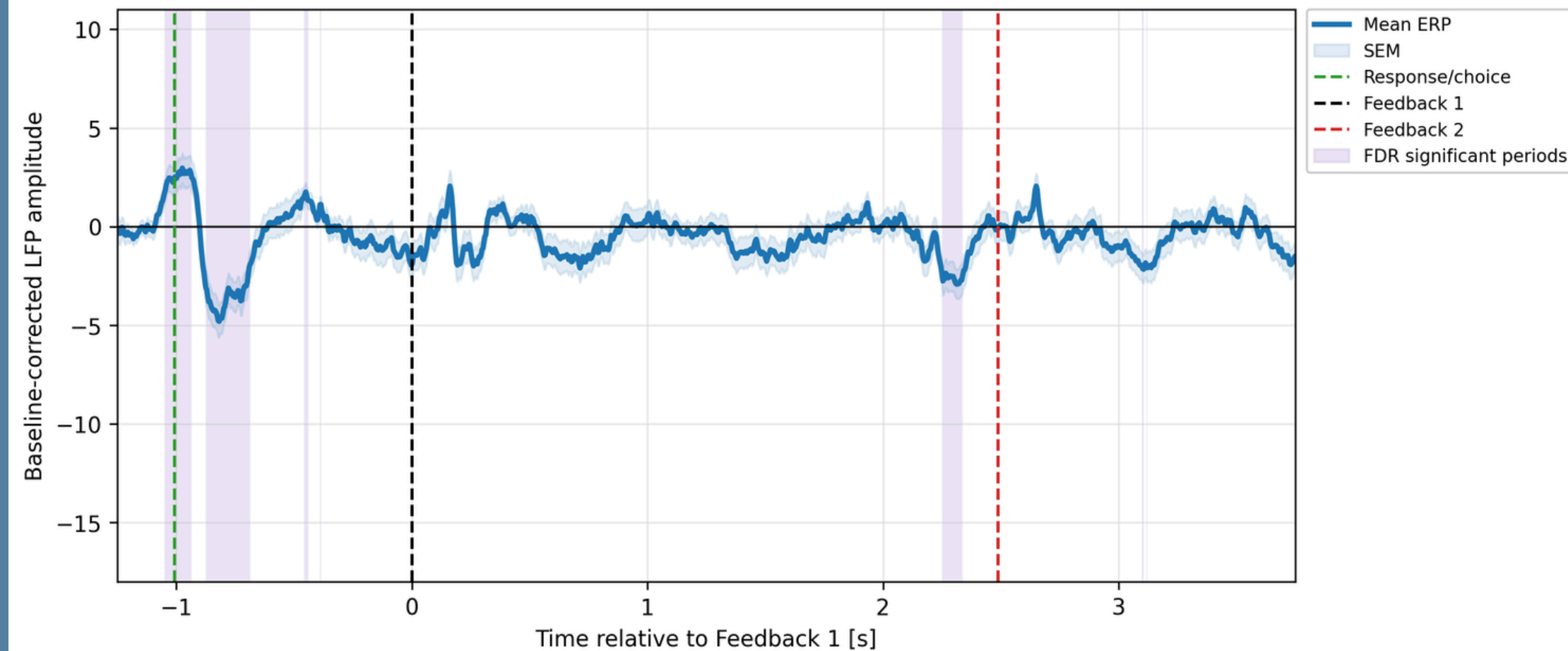
NATALIA  
 $\text{CHI}^2 = 24.276$   
P-VALUE < 0.0001



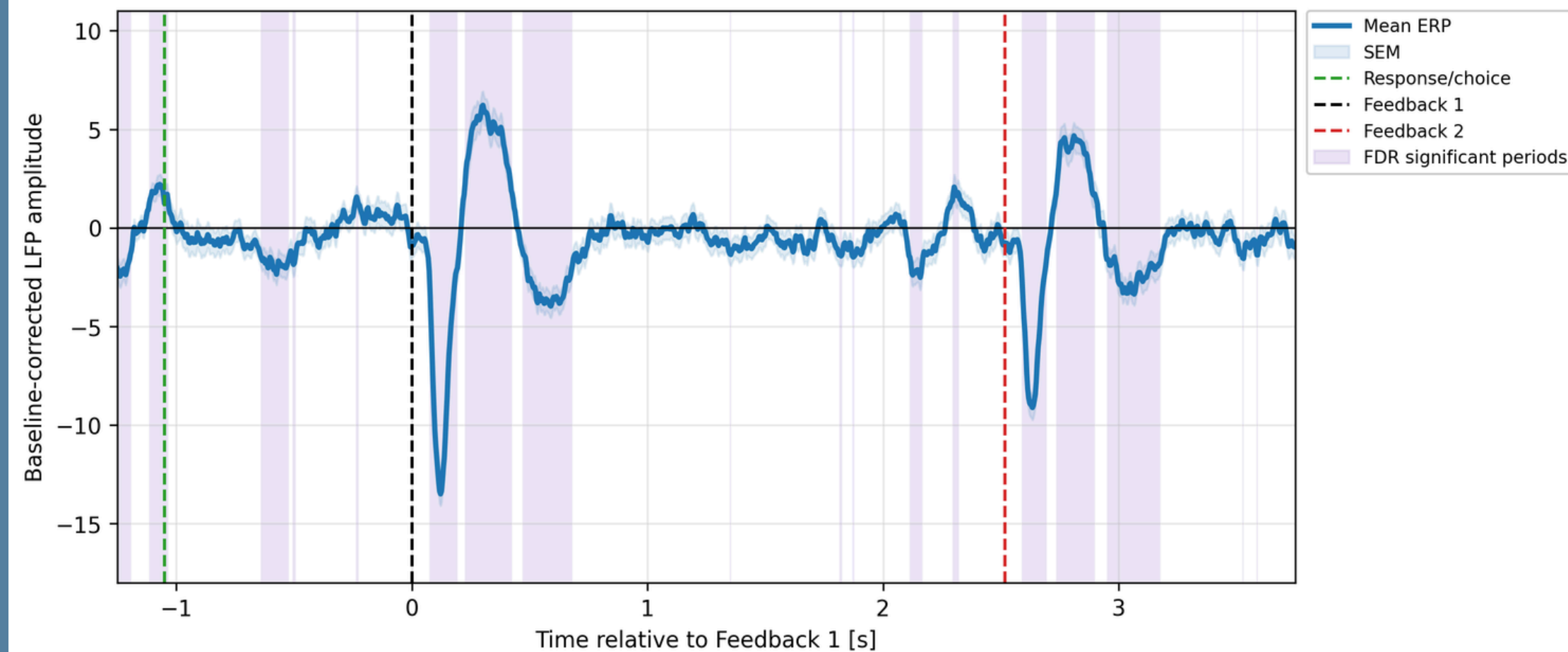
# EVENT-RELATED POTENTIAL

- Natalia showed clear ERP deflections after Feedback 1 and around Feedback 2.
- Julia showed a weaker ERP pattern, with the most visible change occurring around the response/choice period rather than clearly after feedback.
- Purple shaded regions indicate time periods where ERP significantly differed from baseline after Wilcoxon signed-rank test + FDR correction.
- Overall, ERP responses suggest stronger feedback-related LFP modulation in Natalia than in Julia.

Julia: global feedback-locked ERP (n = 150 trials)



Natalia: global feedback-locked ERP (n = 150 trials)

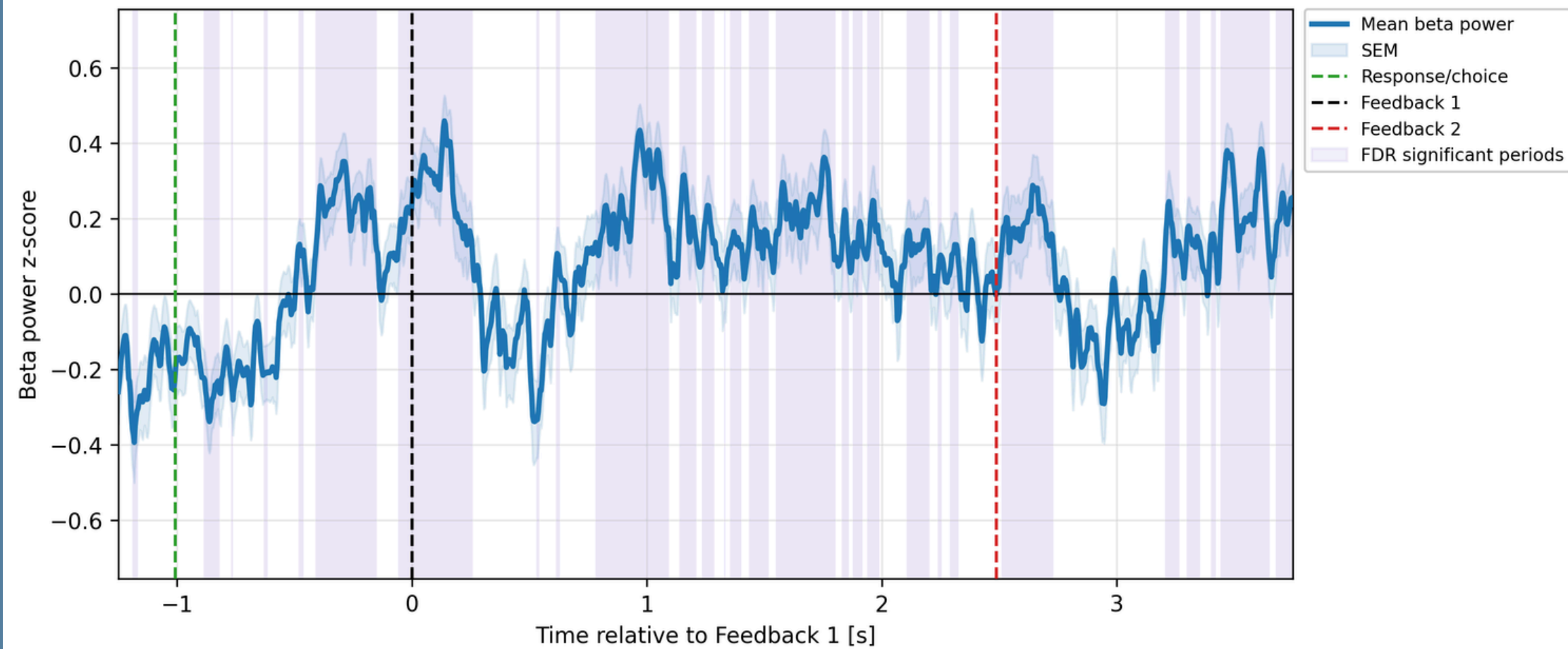




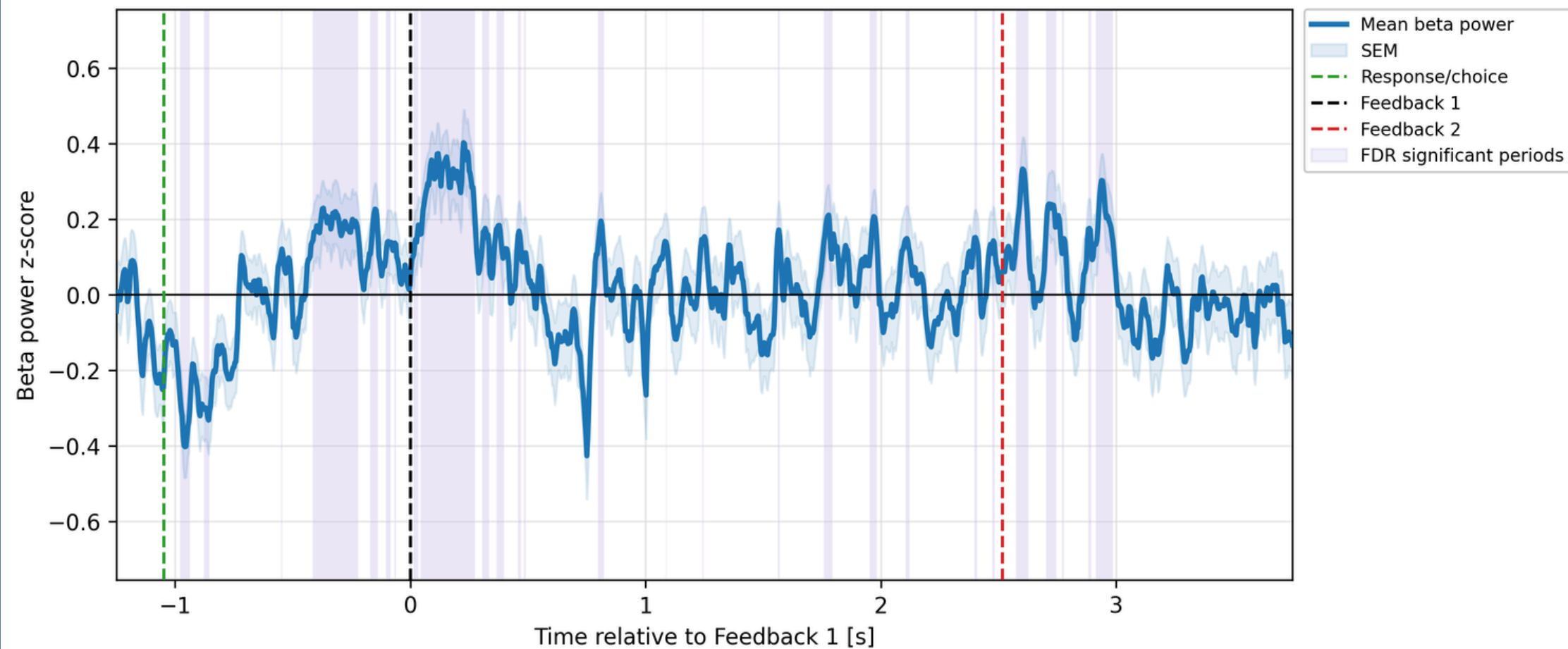
# BETA BAND ANALYSIS

- Julia showed more widespread significant beta periods across time.
- Natalia showed fewer significant periods, but some were more closely aligned with task events.
- Negative beta values around the response/choice period suggest beta desynchronization.
- The timing and strength of beta modulation differed between patients, suggesting patient-specific STN dynamics.

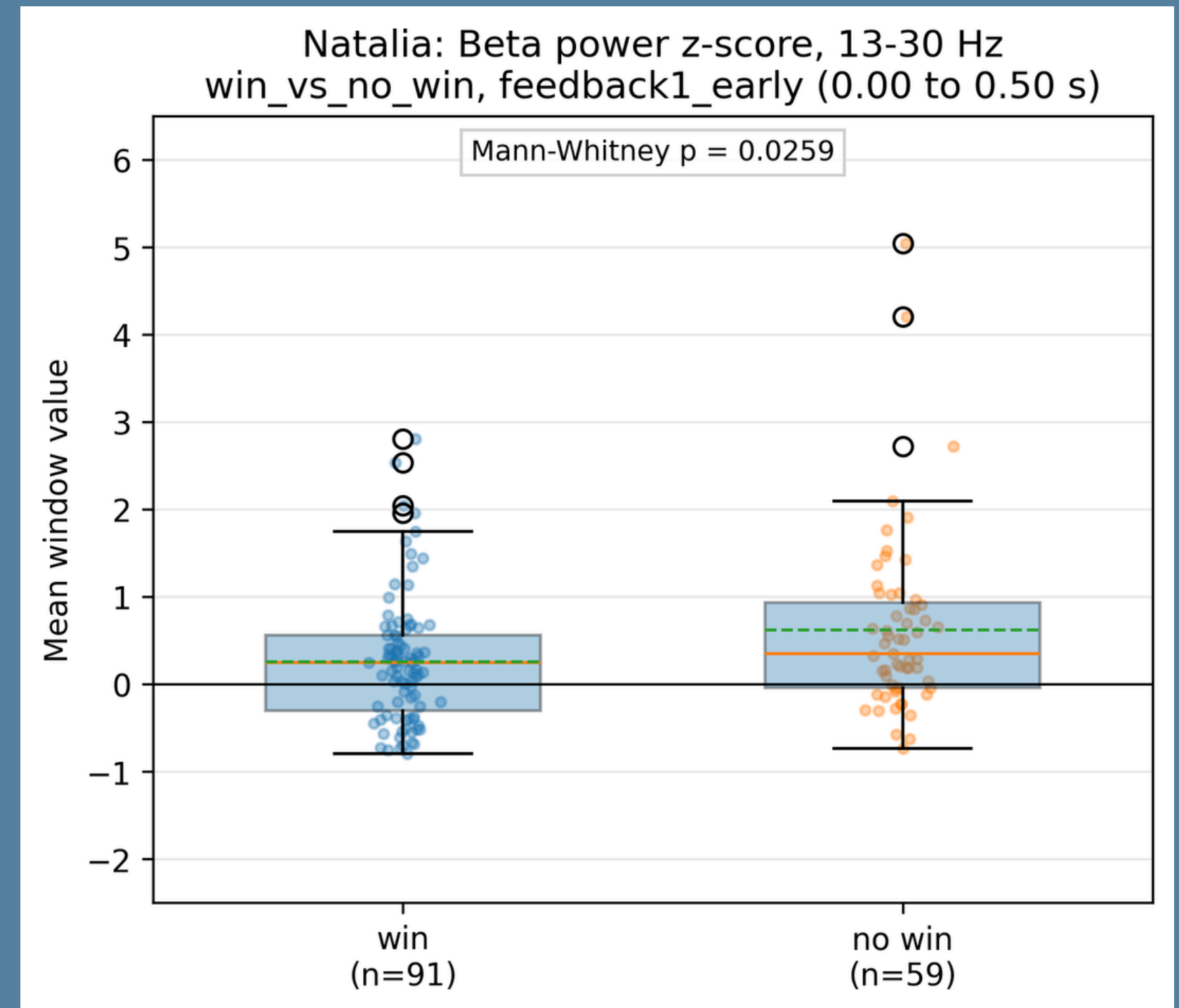
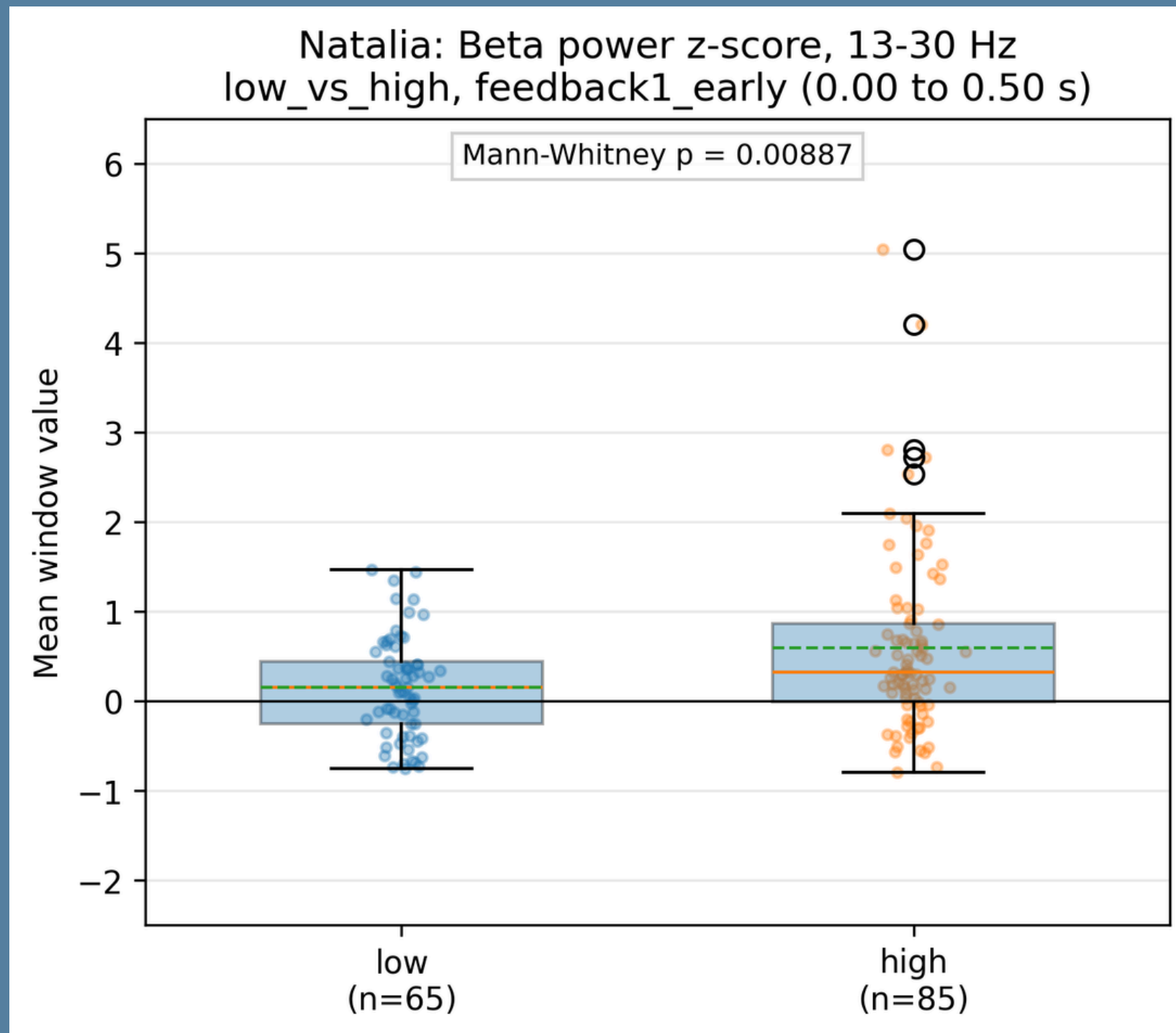
Julia: global feedback-locked beta power (n = 150 trials)



Natalia: global feedback-locked beta power (n = 150 trials)



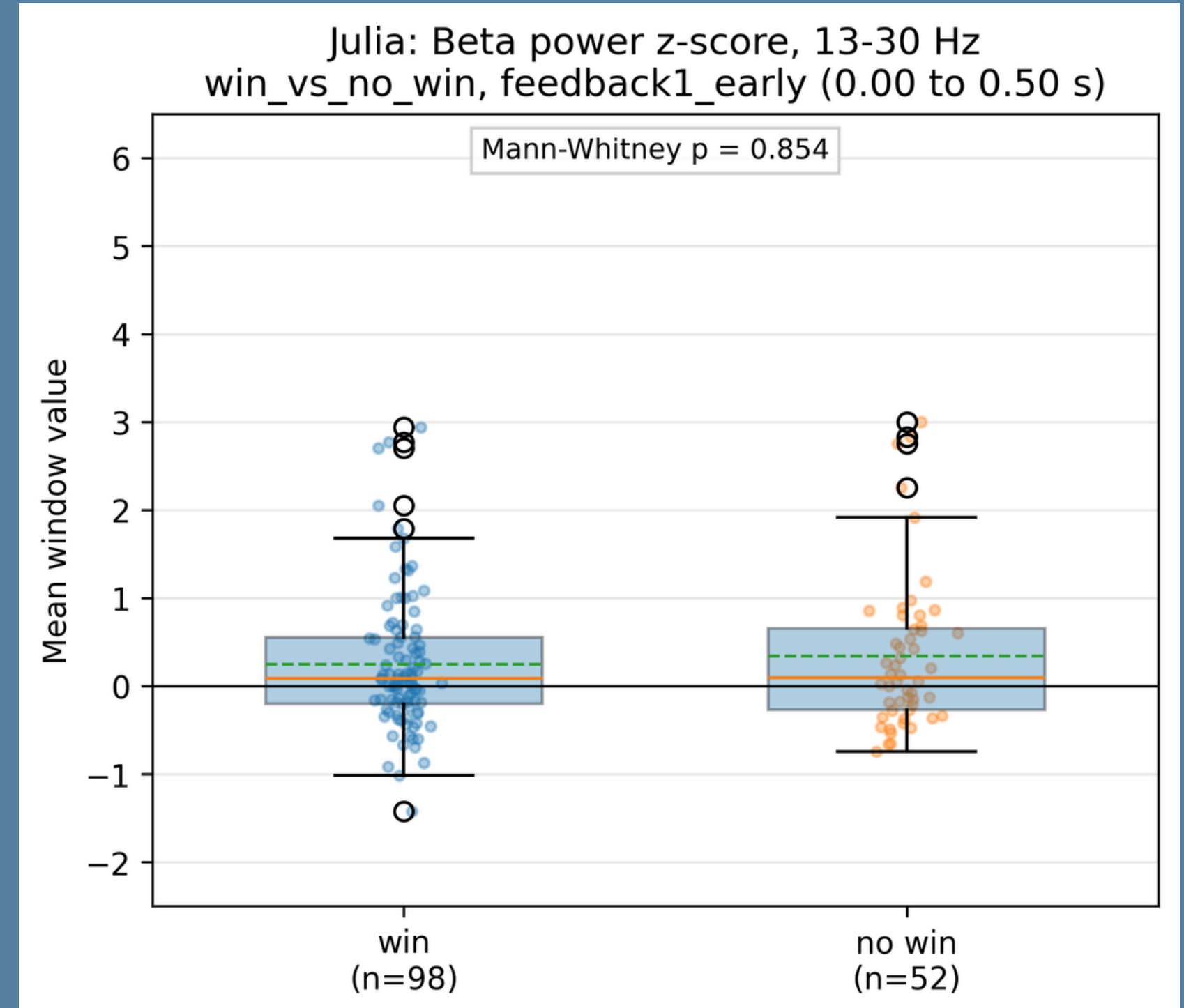
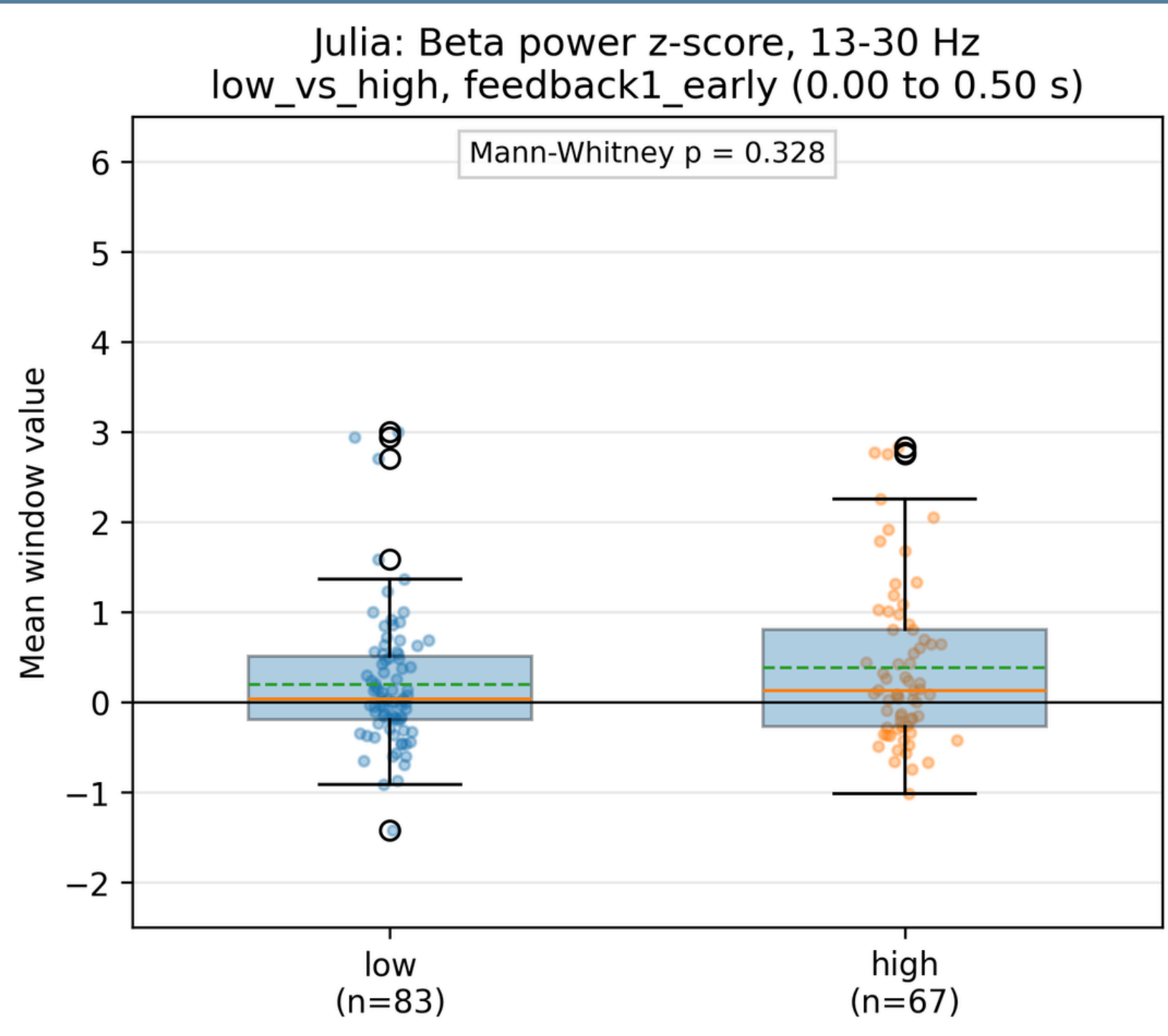
# NATALIA – BETA CONDITION COMPARISON



Natalia showed significant condition-related beta differences in the early Feedback 1 window. Beta power was higher for high-risk than low-risk trials and higher for no-win than win trials, suggesting that beta activity may reflect risk and outcome processing after feedback.



# JULIA– BETA CONDITION COMPARISON



Julia's beta response after Feedback 1 was not significantly modulated by risk level or feedback outcome.

## MAIN CONCLUSIONS – BEHAVIOURAL ANALYSIS

- Reaction times differed by chosen stimuli, but effects were inconsistent across patients
- Julia's patient: significant strategy change in Block 2
- Natalia's patient: gradual increase in risk-taking, but not significant
- Strong significant effect of previous feedback on next choice, showing history-dependent decision-making
- Overall: behavior is adaptive, feedback-driven, and individually variable

## CONCLUSION – LFP ANALYSIS

- ERP and beta analyses showed task-related neural modulation around response and feedback events.
- Natalia showed clearer ERP responses, while Julia showed more variable activity.
- Condition comparisons showed exploratory beta trends
- The results suggest possible patient-specific STN activity patterns.

THANK  
YOU

